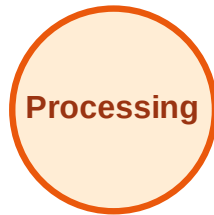


# BFS Traversal

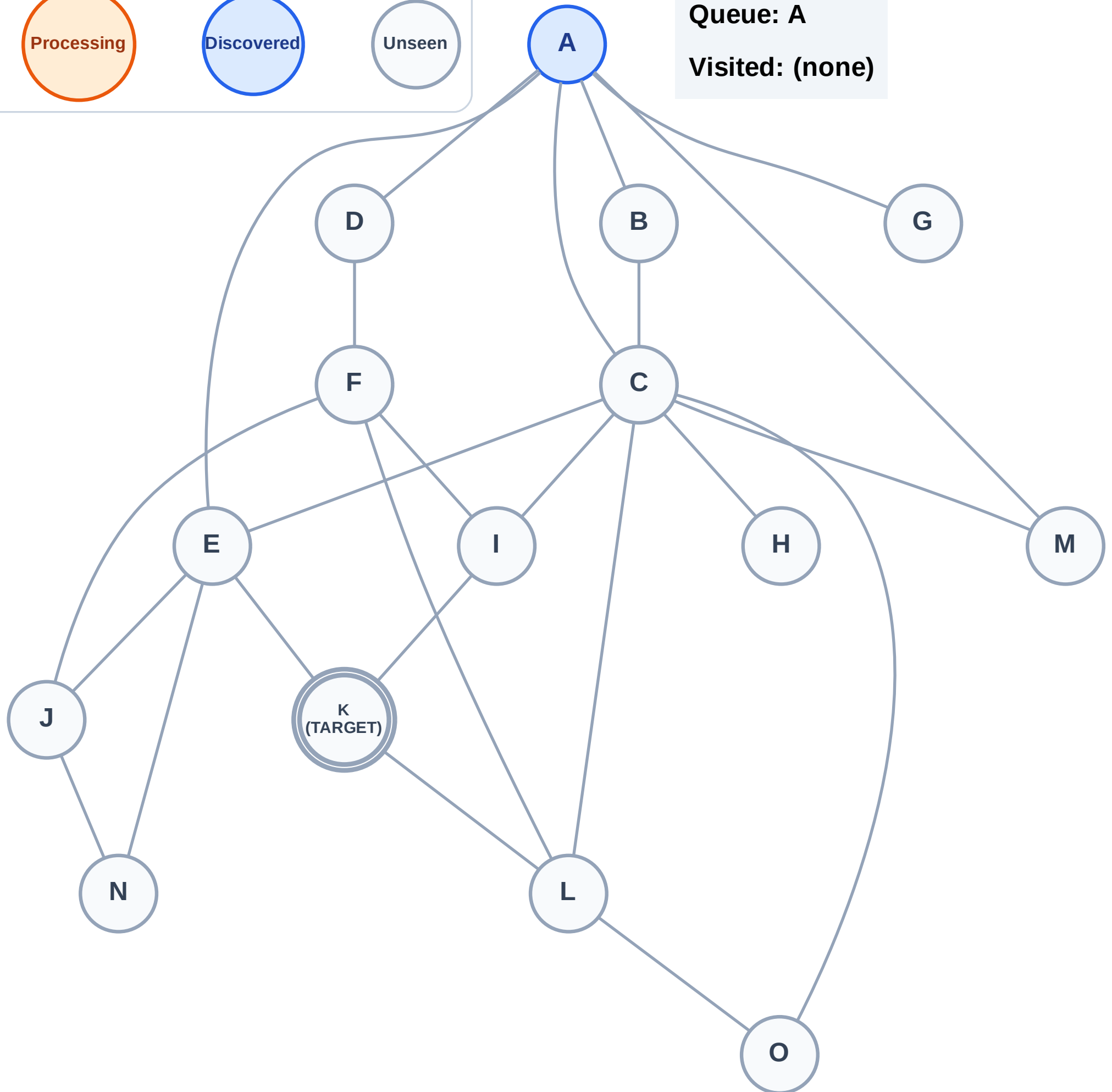
Step 1: Start at A (Seeking target node K)

## Legend



Queue: A

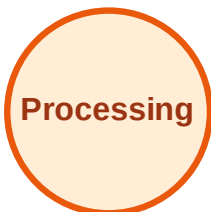
Visited: (none)



# BFS Traversal

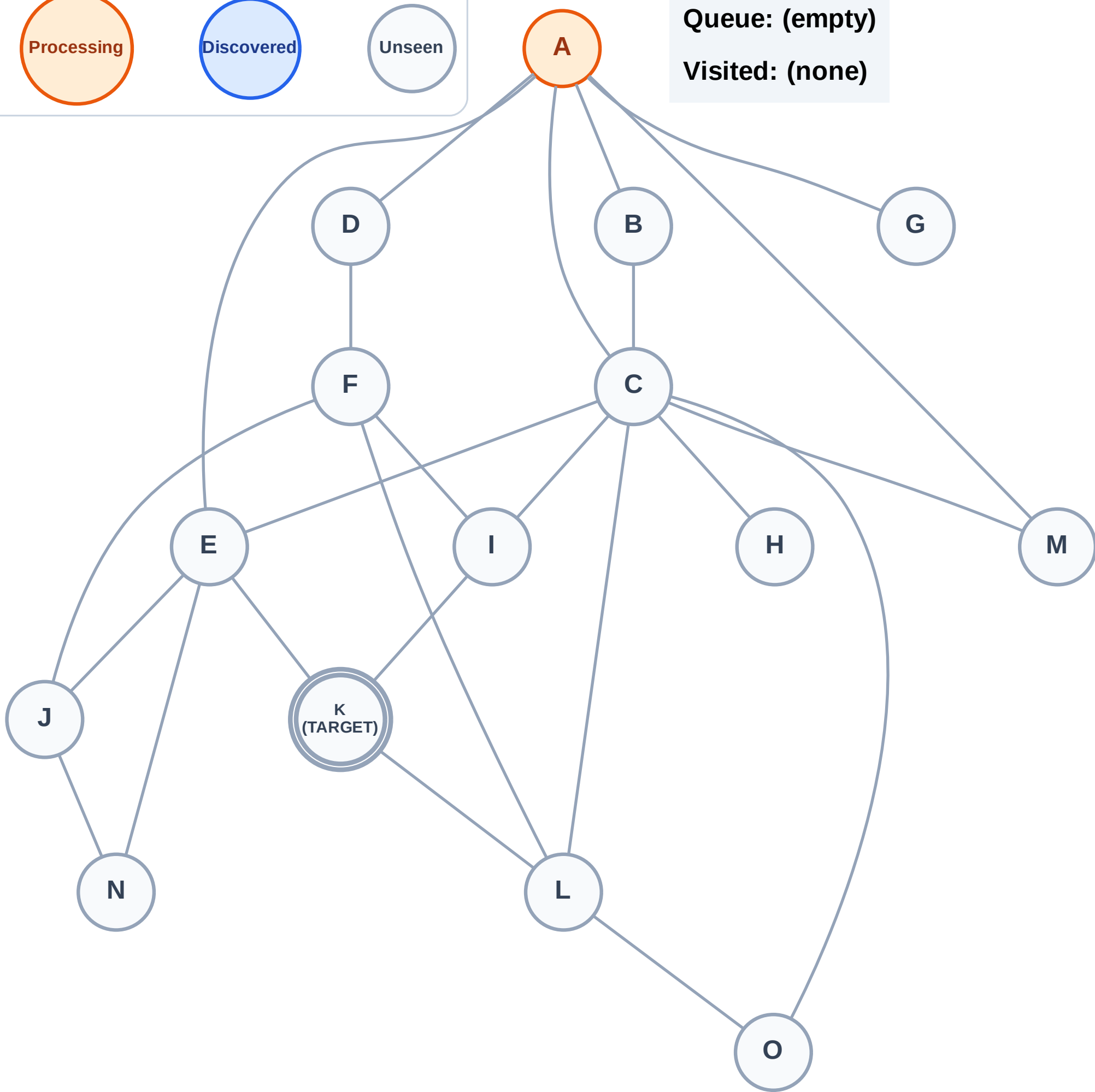
Step 2: Process node A

## Legend



Queue: (empty)

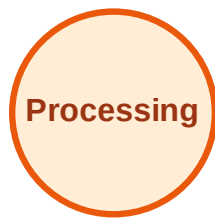
Visited: (none)



# BFS Traversal

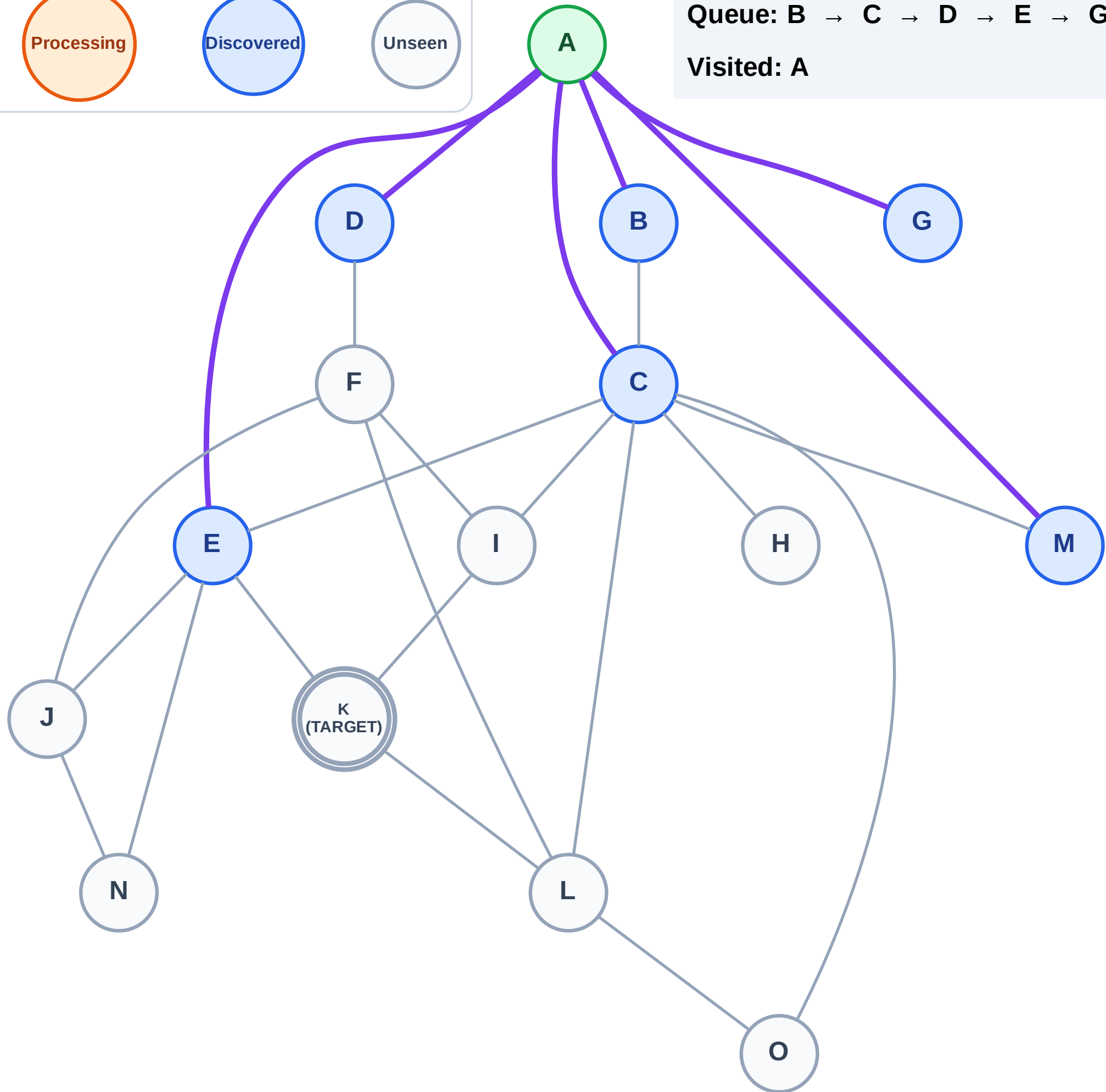
Step 3: Discover B, C, D, E, G, M from A

## Legend



Queue: B → C → D → E → G → M

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

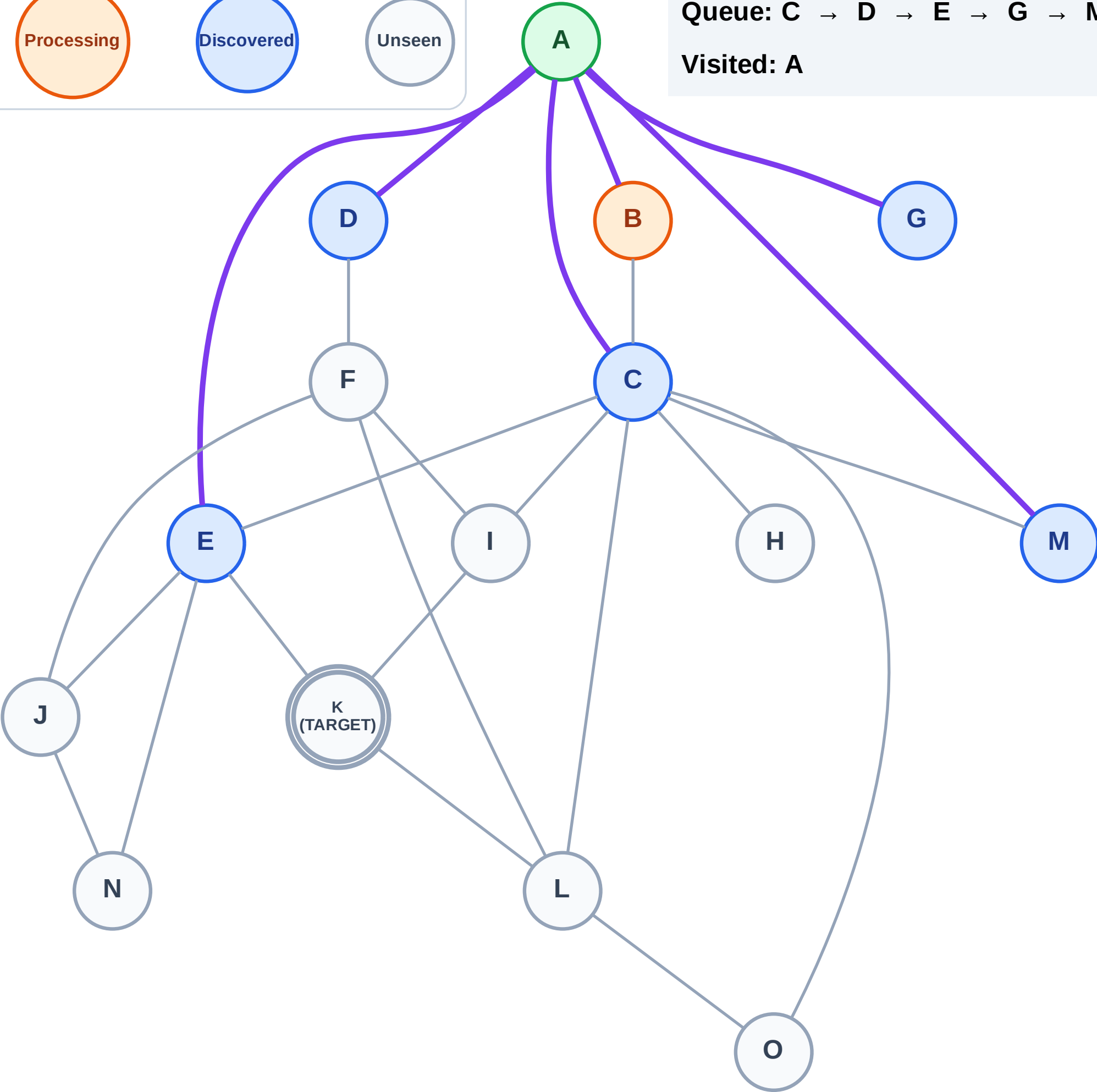
Processing

Discovered

Unseen

Queue: C → D → E → G → M

Visited: A



# BFS Traversal

Step 5: No new neighbors from B

Legend

Complete

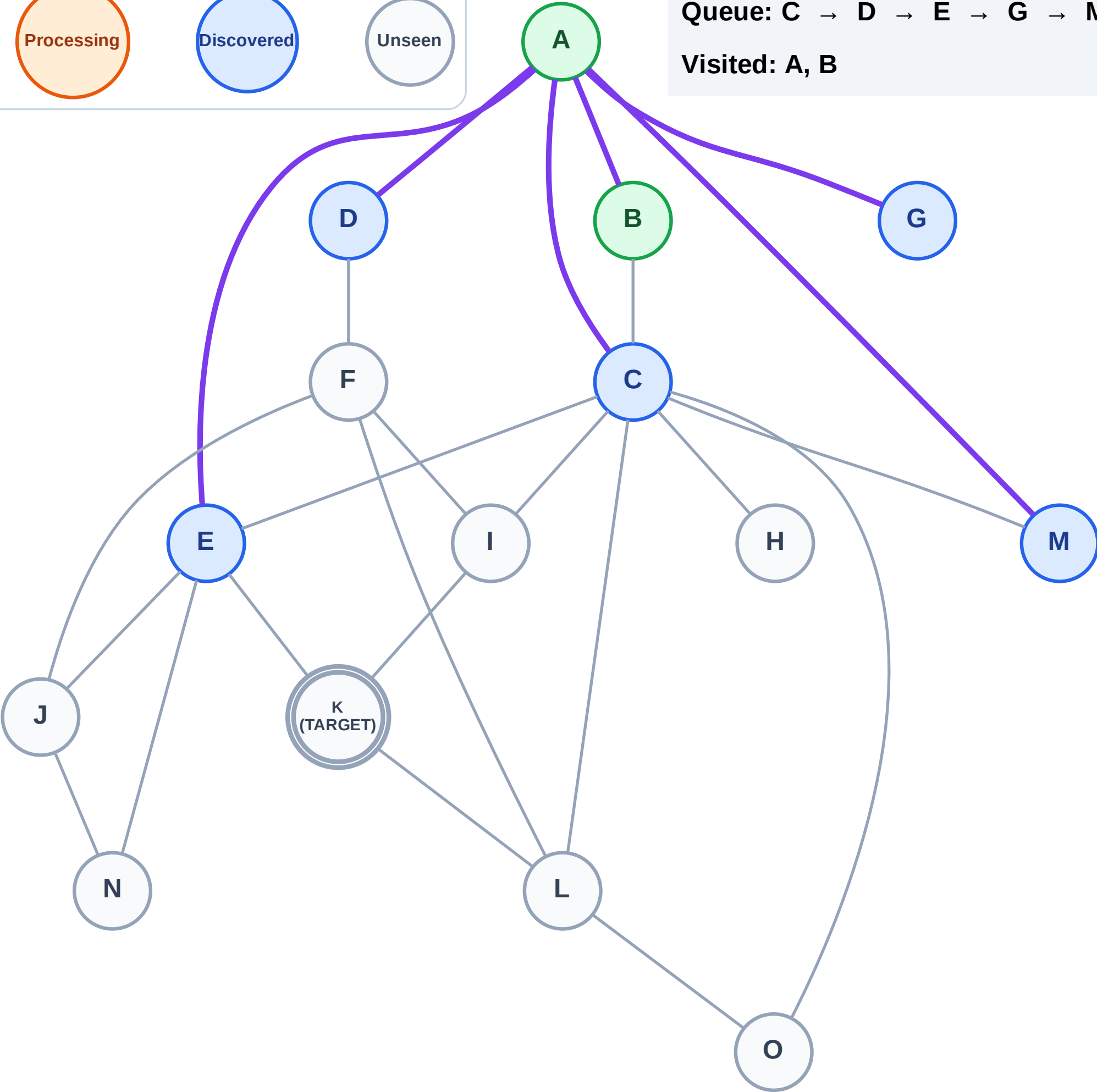
Processing

Discovered

Unseen

Queue: C → D → E → G → M

Visited: A, B



# BFS Traversal

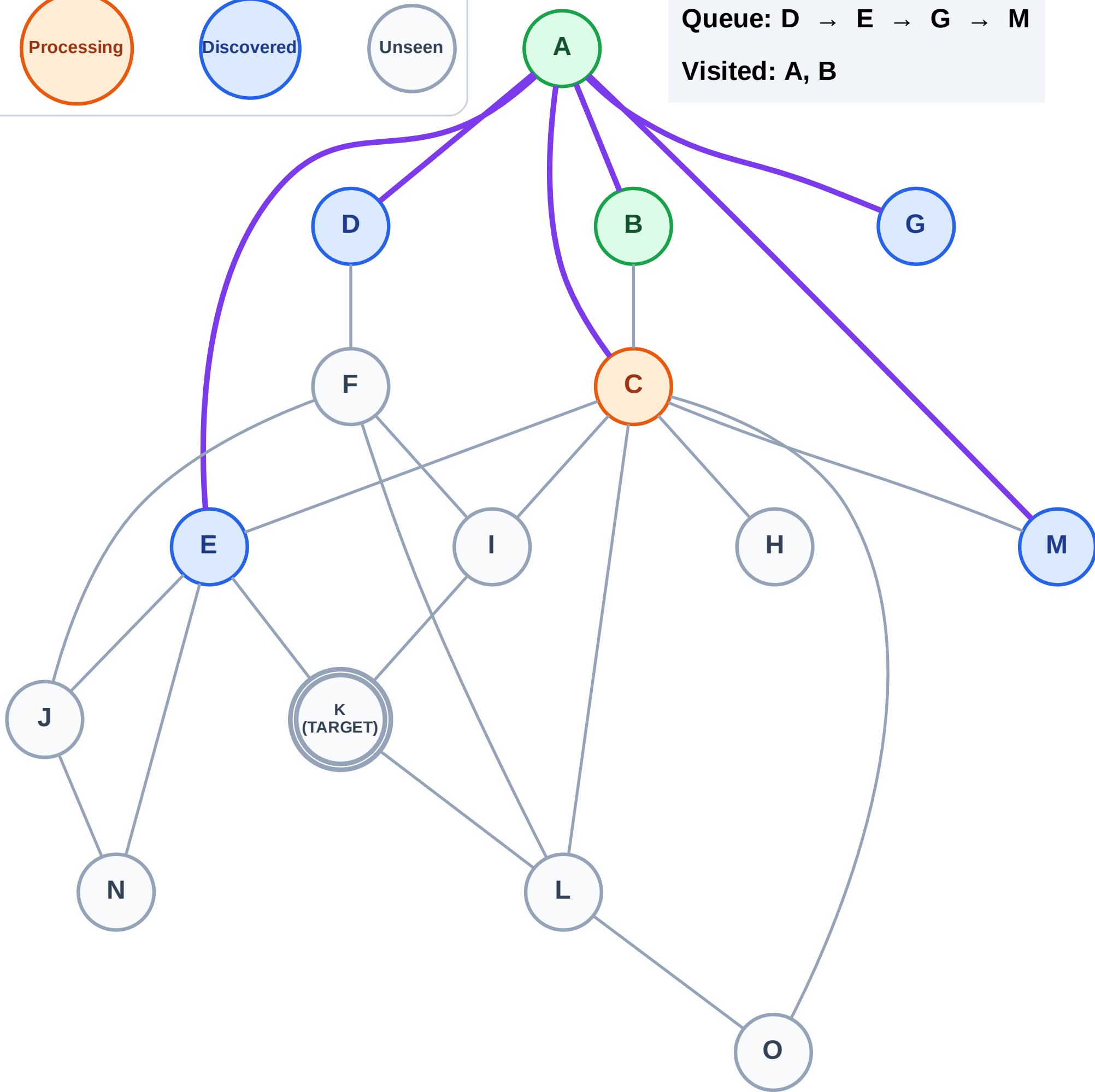
Step 6: Process node C

## Legend



Queue: D → E → G → M

Visited: A, B



# BFS Traversal

Step 7: Discover H, I, L, O from C

Legend

Complete

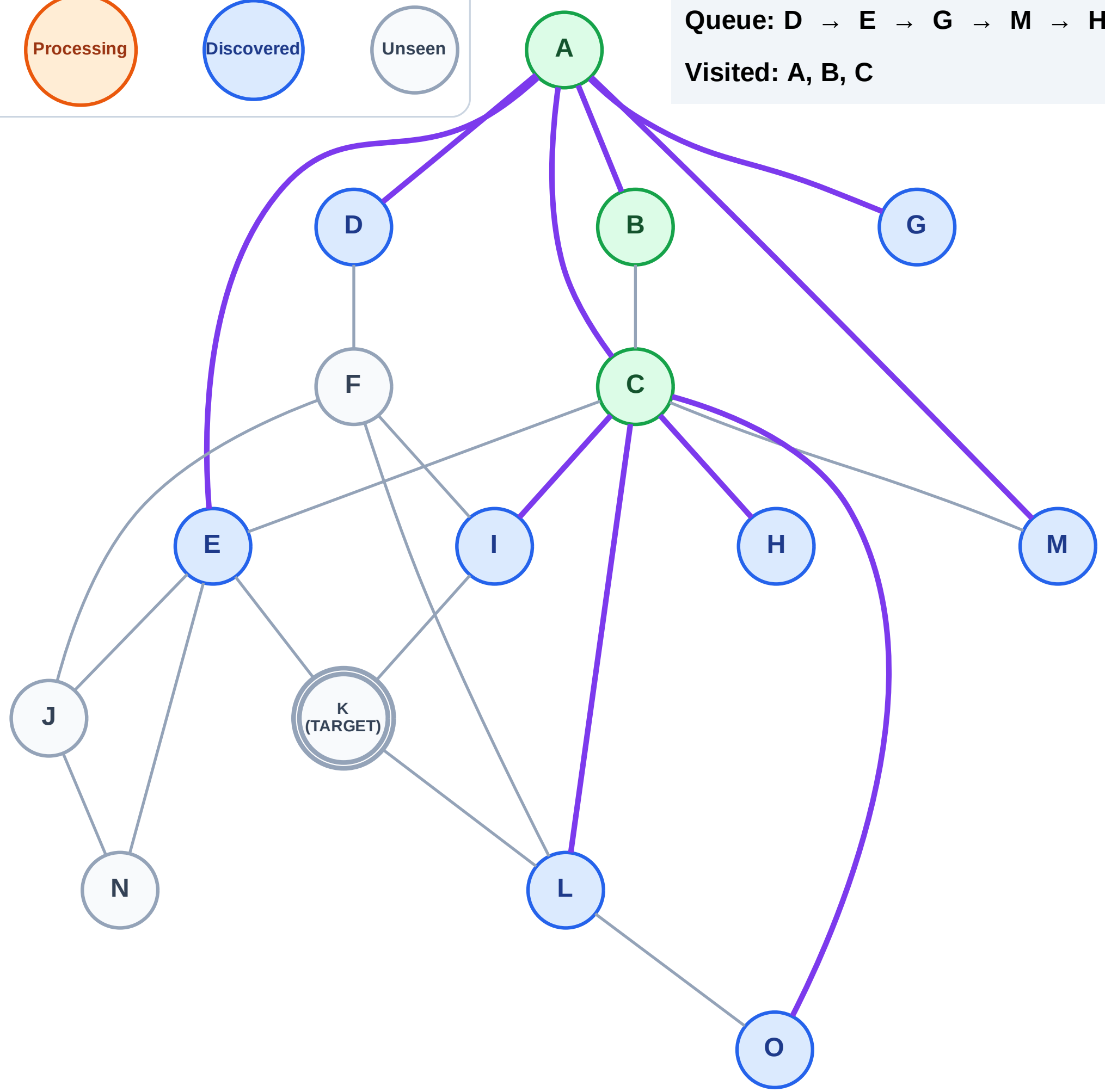
Processing

Discovered

Unseen

Queue: D → E → G → M → H → I → L → O

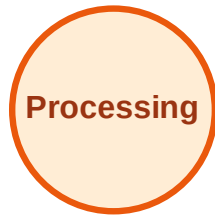
Visited: A, B, C



# BFS Traversal

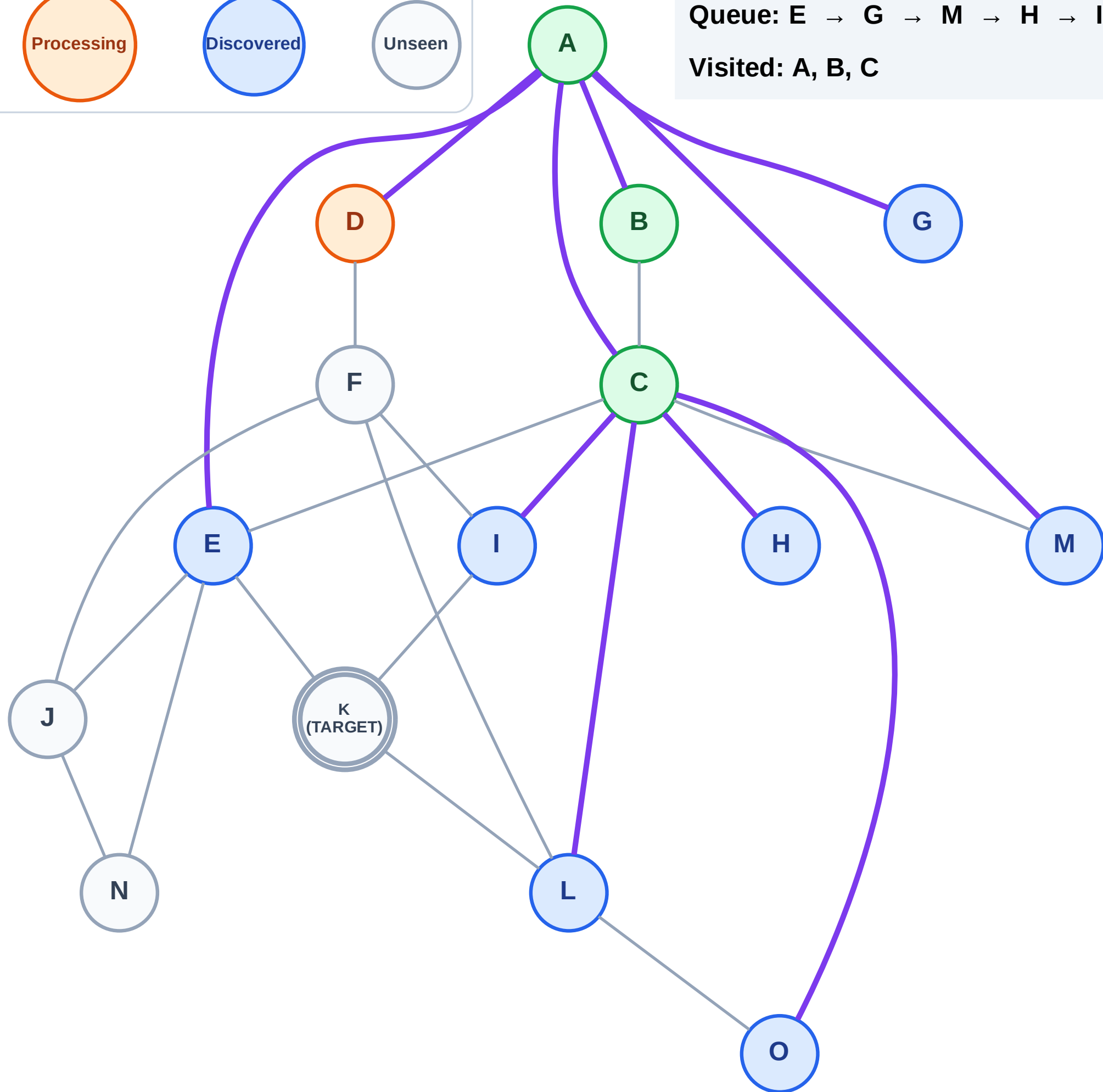
Step 8: Process node D

## Legend



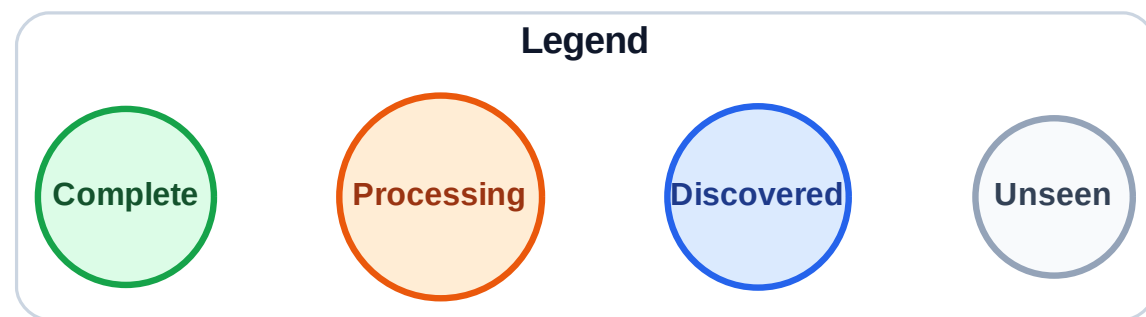
Queue: E → G → M → H → I → L → O

Visited: A, B, C



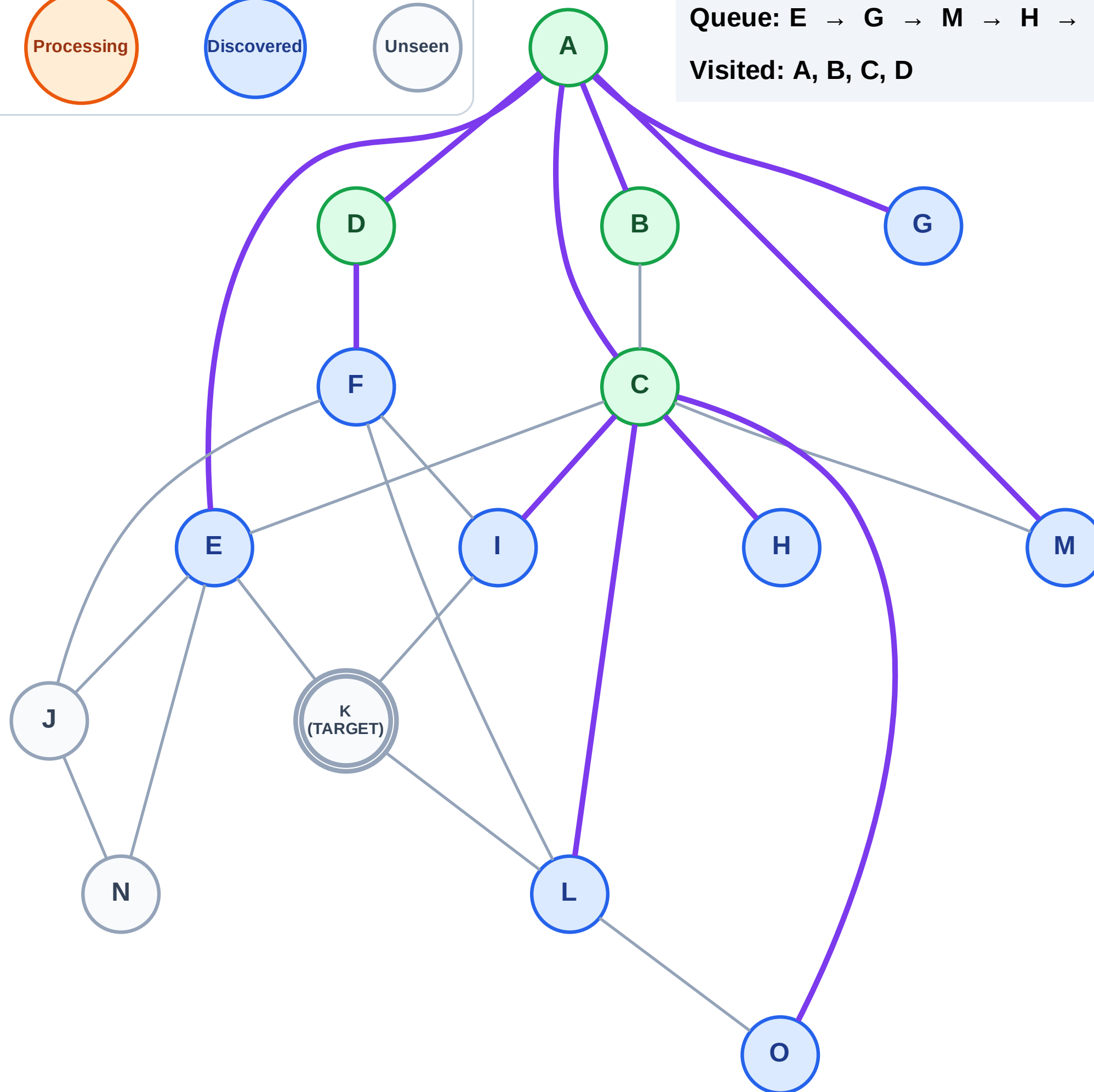
# BFS Traversal

Step 9: Discover F from D



Queue: E → G → M → H → I → L → O → F

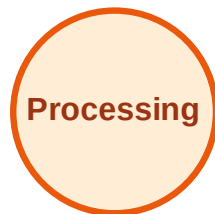
Visited: A, B, C, D



# BFS Traversal

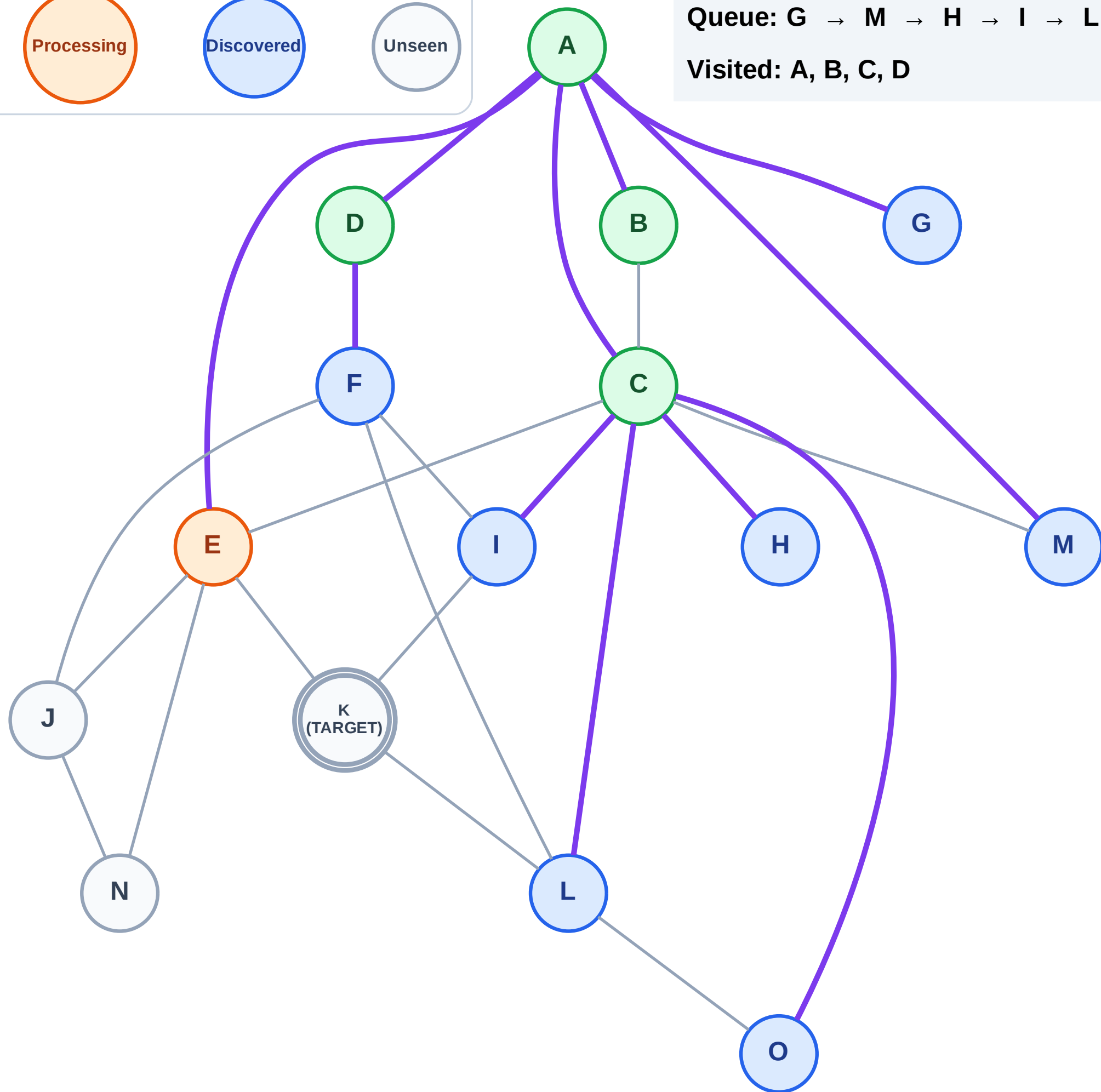
Step 10: Process node E

## Legend



Queue: G → M → H → I → L → O → F

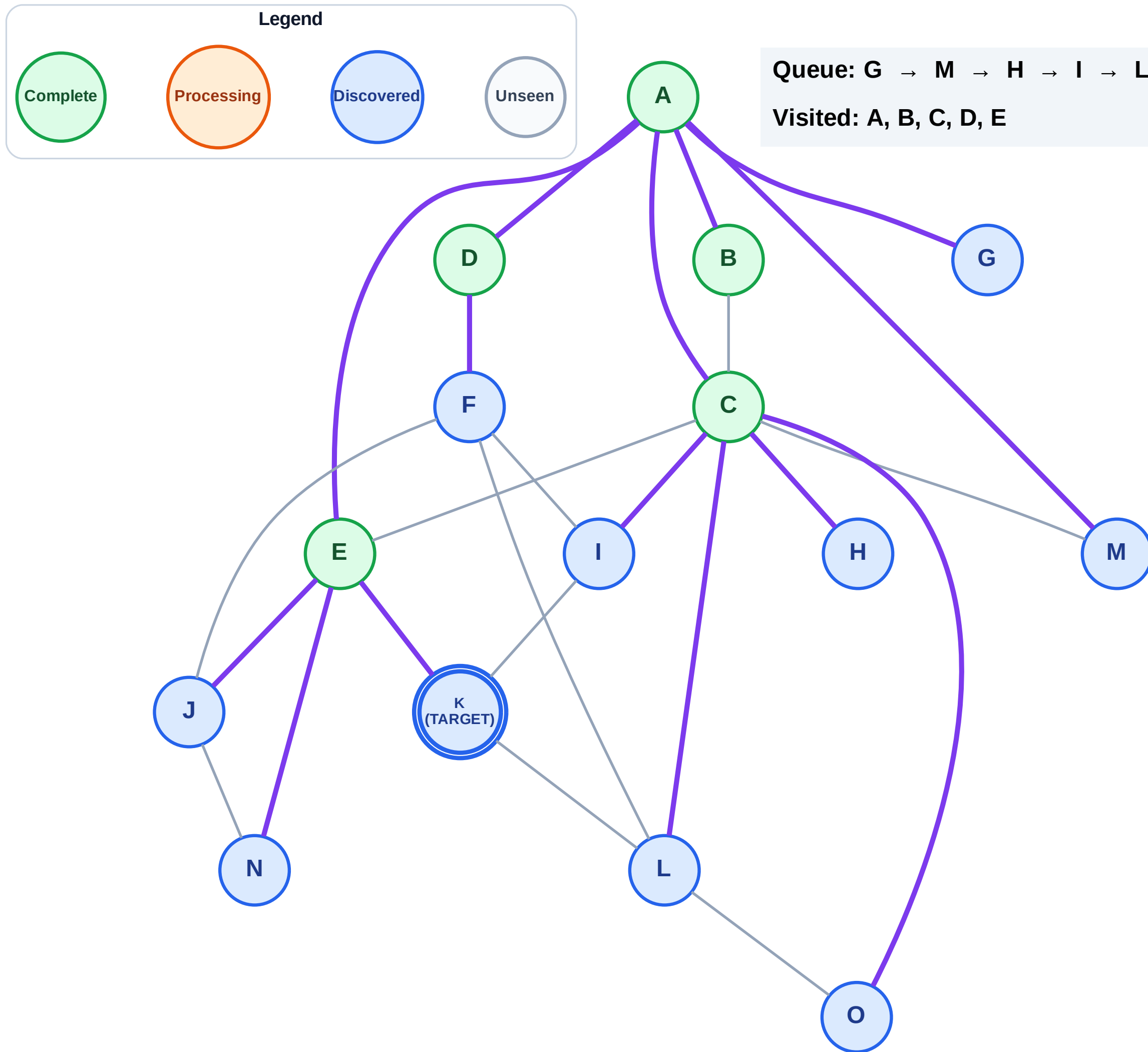
Visited: A, B, C, D



### Step 11: Discover J, K, N from E

**Queue: G → M → H → I → L → O → F → J → K → N**

**Visited: A, B, C, D, E**



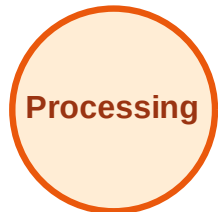
# BFS Traversal

Step 12: Process node G

## Legend



Complete



Processing



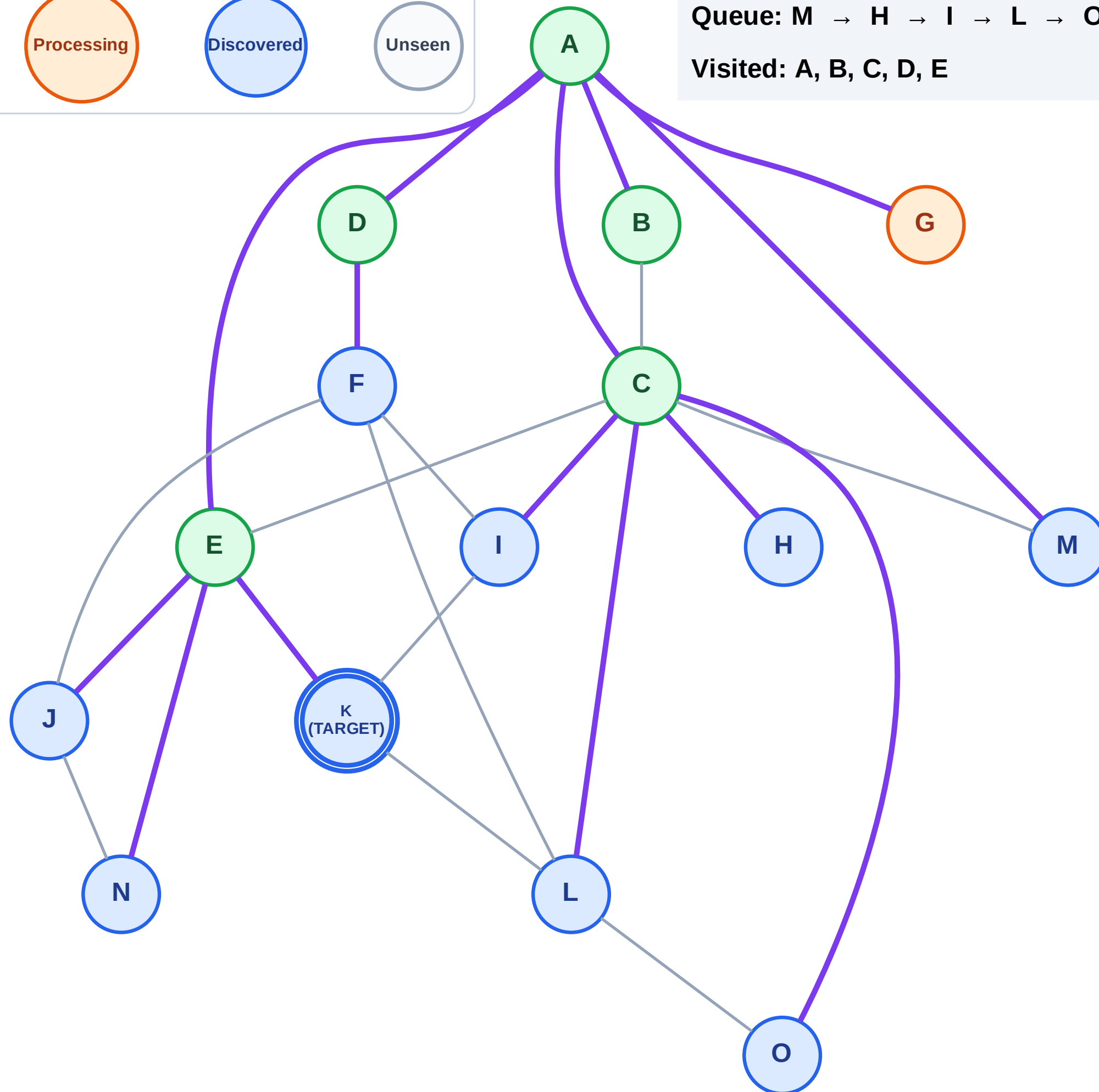
Discovered



Unseen

Queue: M → H → I → L → O → F → J → K → N

Visited: A, B, C, D, E



# BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

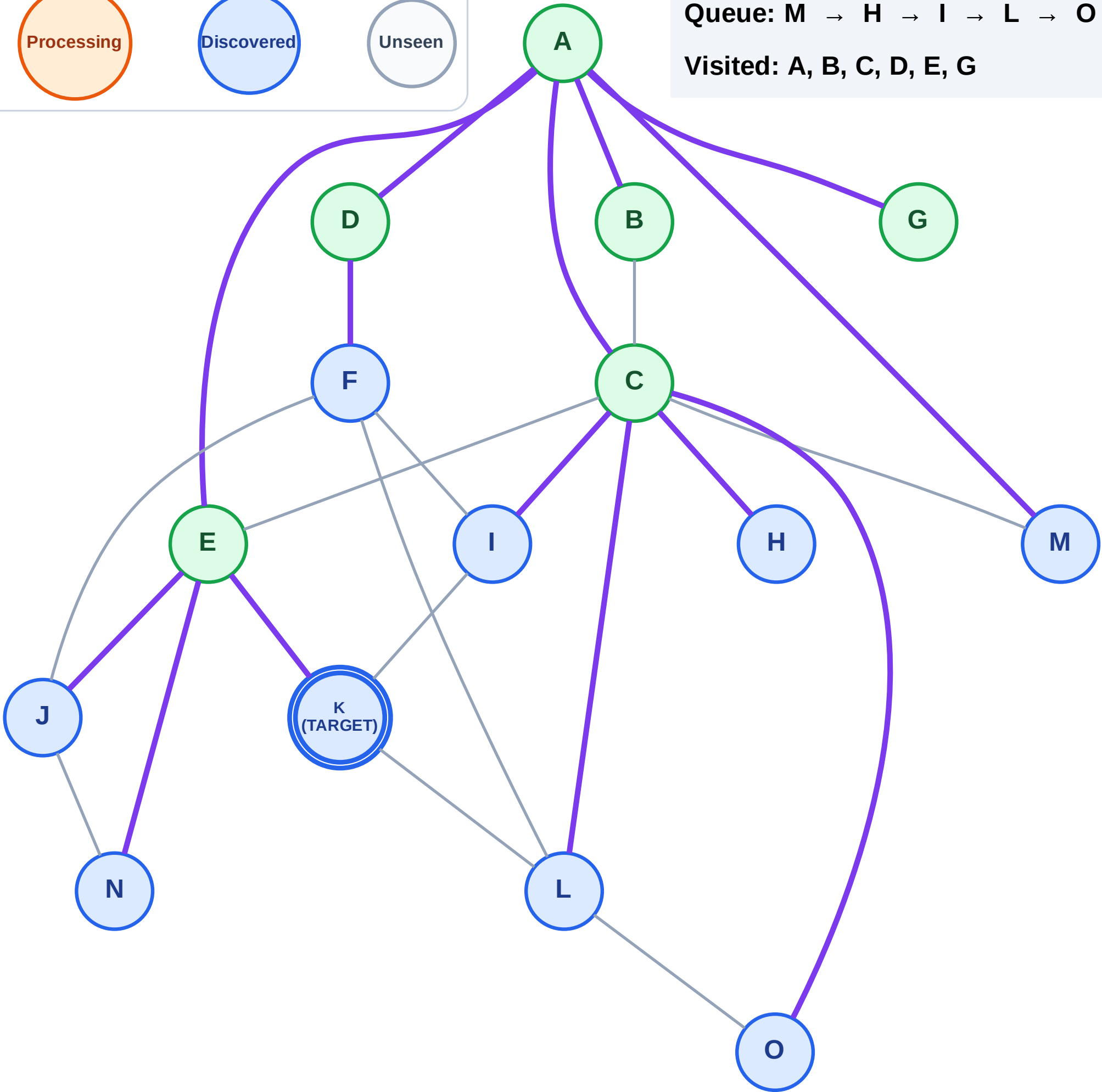
Processing

Discovered

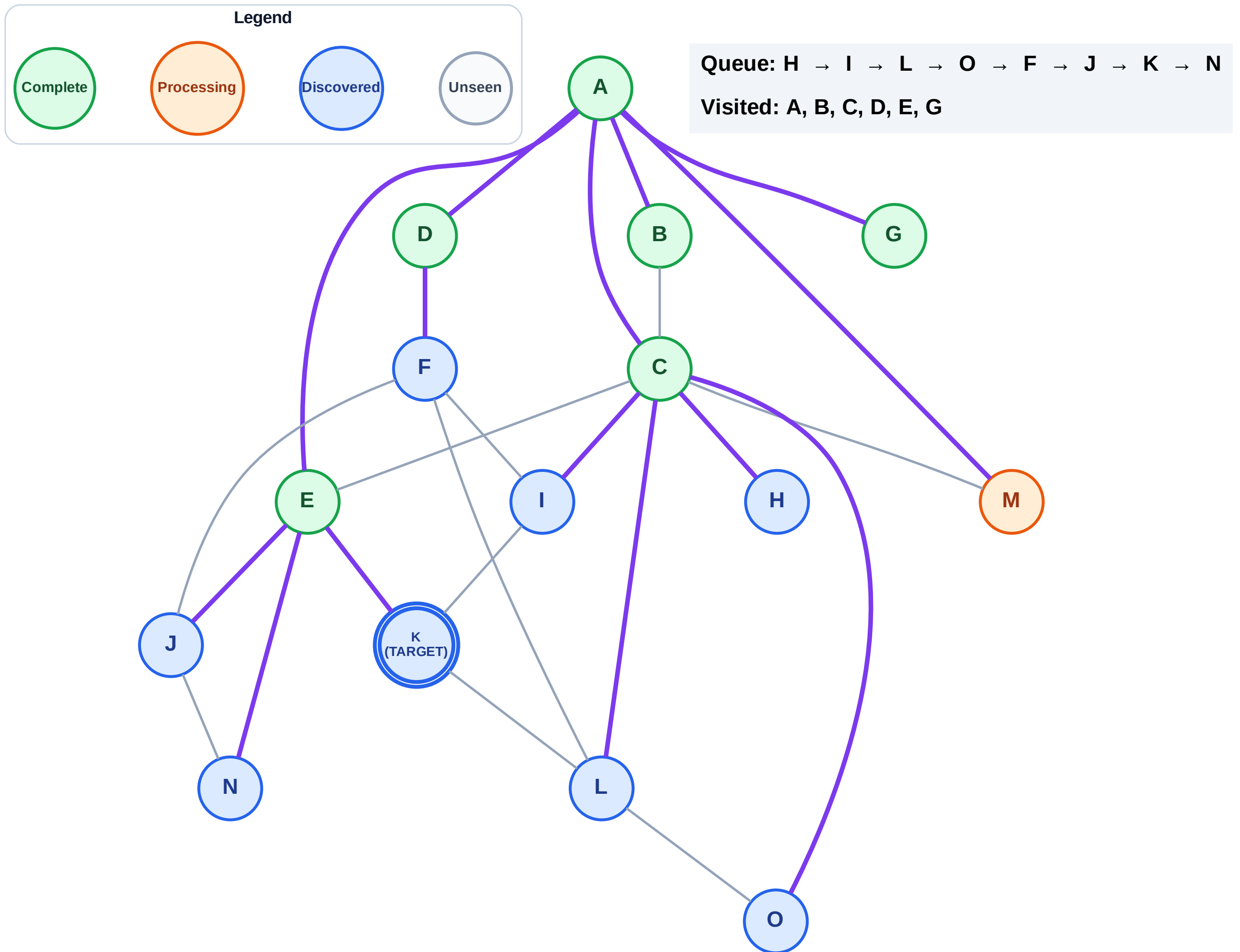
Unseen

Queue: M → H → I → L → O → F → J → K → N

Visited: A, B, C, D, E, G



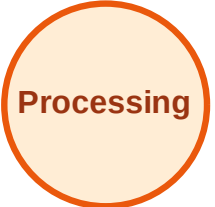
## Step 14: Process node M



# BFS Traversal

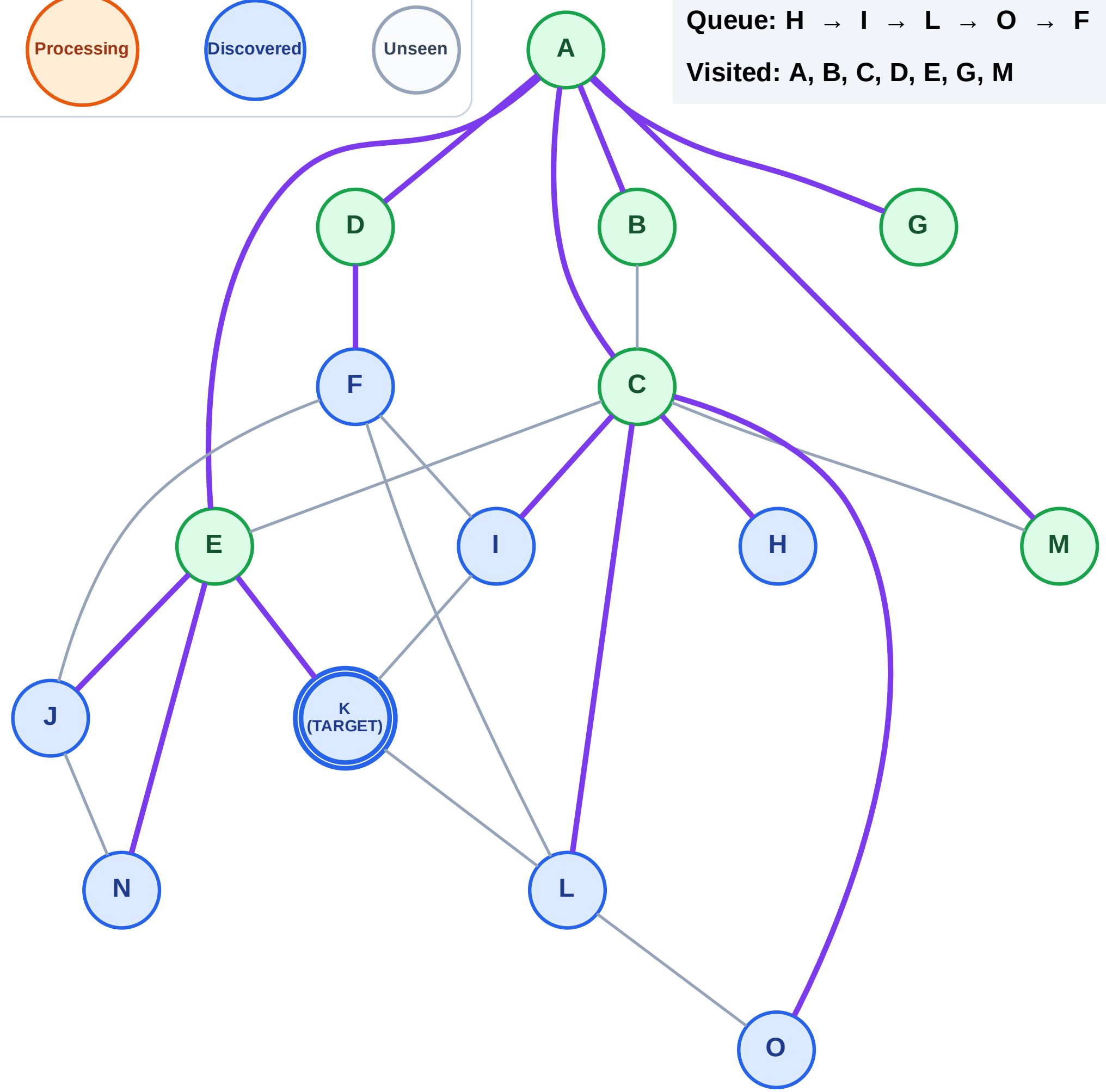
Step 15: No new neighbors from M

## Legend



Queue: H → I → L → O → F → J → K → N

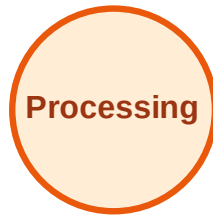
Visited: A, B, C, D, E, G, M



# BFS Traversal

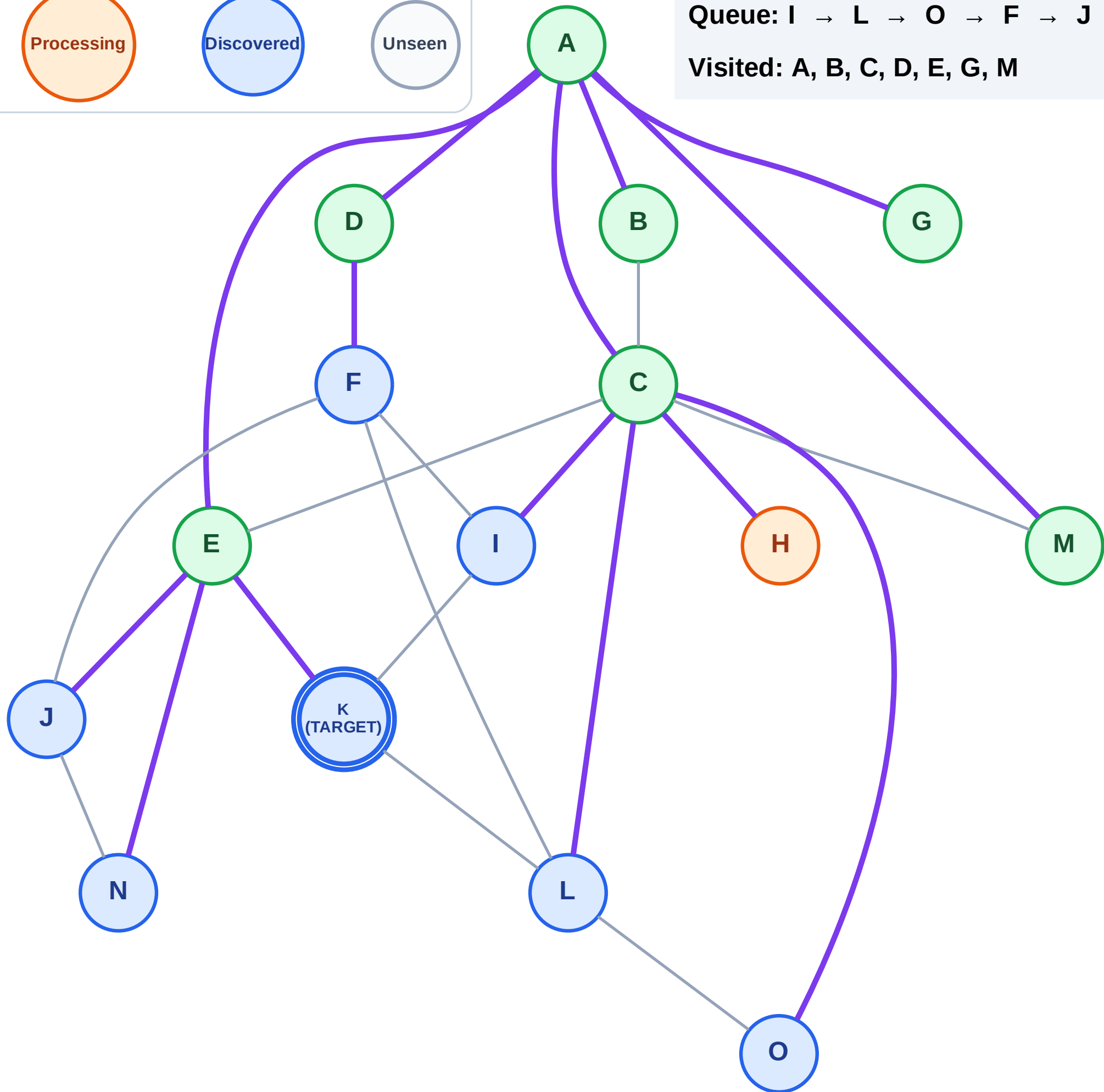
Step 16: Process node H

## Legend



Queue: I → L → O → F → J → K → N

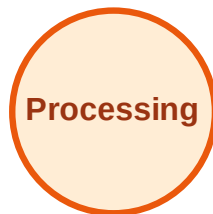
Visited: A, B, C, D, E, G, M



# BFS Traversal

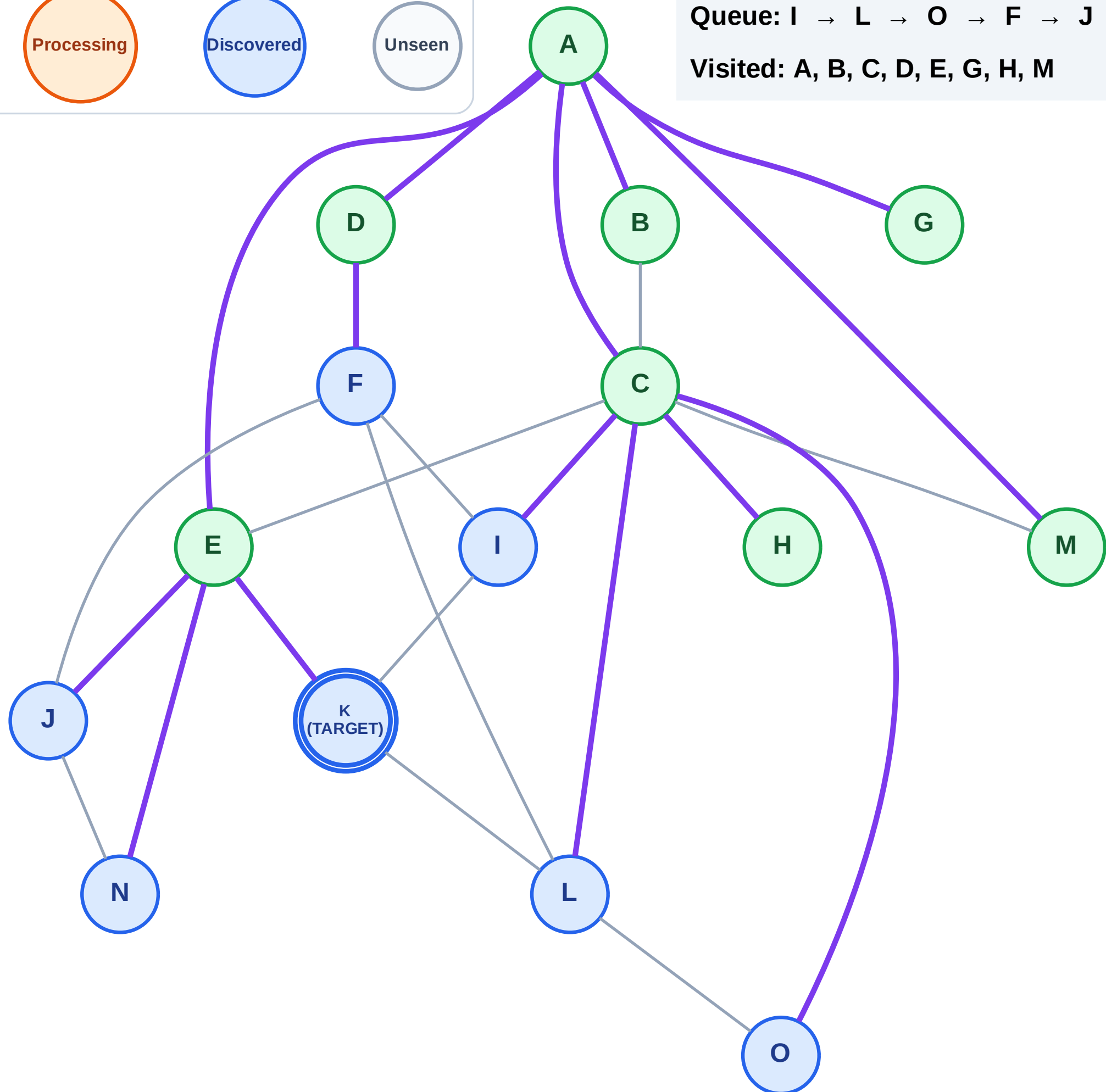
Step 17: No new neighbors from H

## Legend



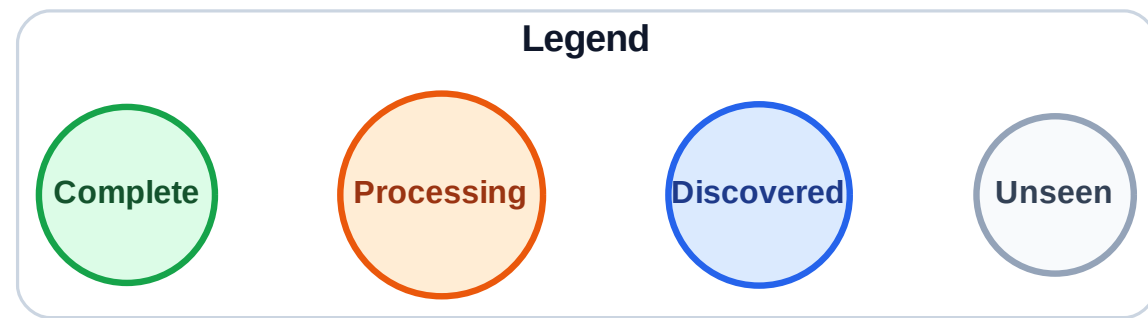
Queue: I → L → O → F → J → K → N

Visited: A, B, C, D, E, G, H, M



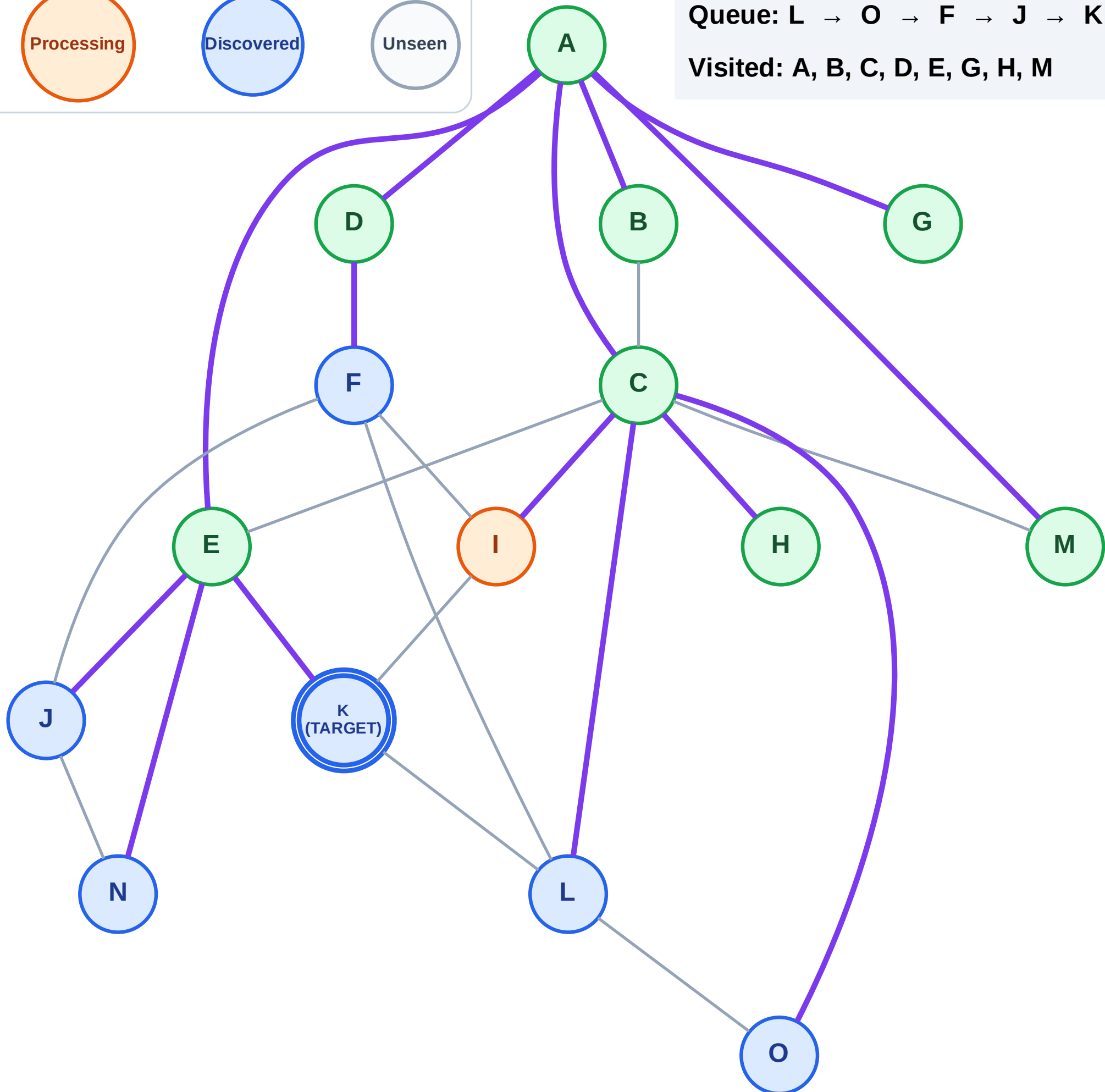
# BFS Traversal

Step 18: Process node I



Queue: L → O → F → J → K → N

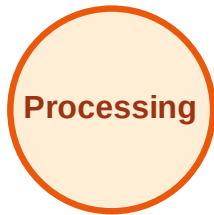
Visited: A, B, C, D, E, G, H, M



# BFS Traversal

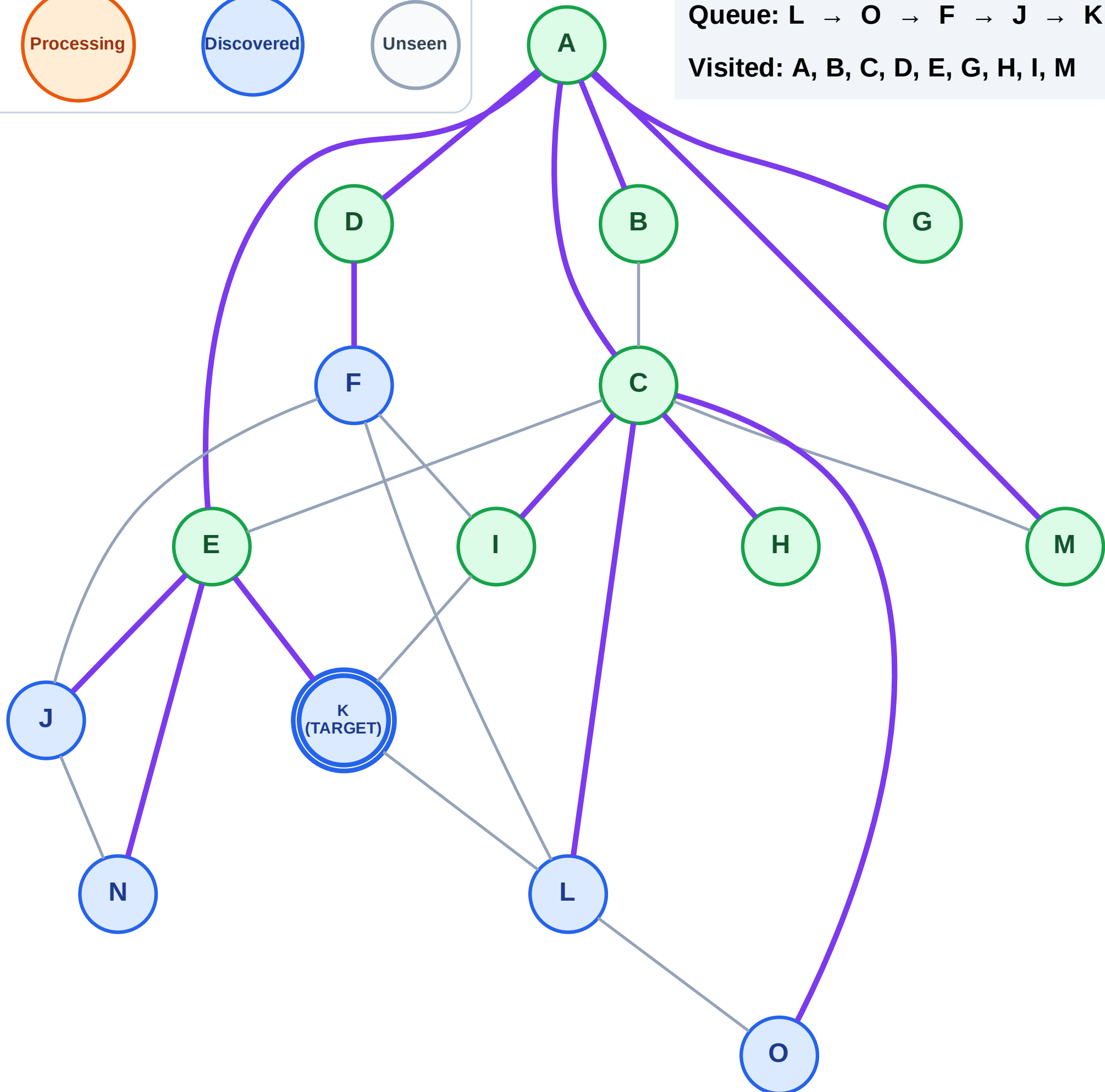
Step 19: No new neighbors from I

## Legend



Queue: L → O → F → J → K → N

Visited: A, B, C, D, E, G, H, I, M



# BFS Traversal

Step 20: Process node L

## Legend



Complete



Processing



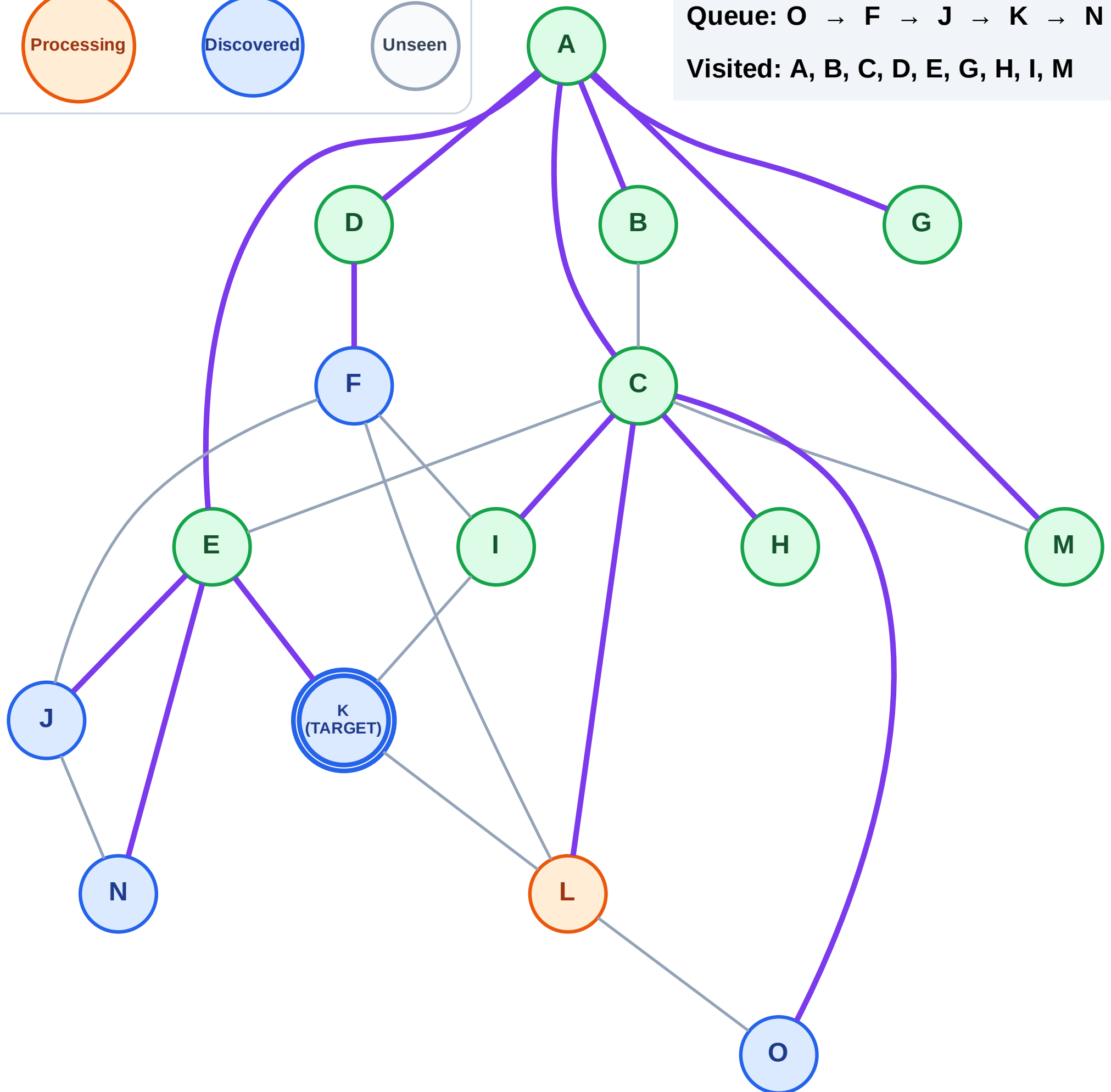
Discovered



Unseen

Queue: O → F → J → K → N

Visited: A, B, C, D, E, G, H, I, M



# BFS Traversal

Step 21: No new neighbors from L

Legend

Complete

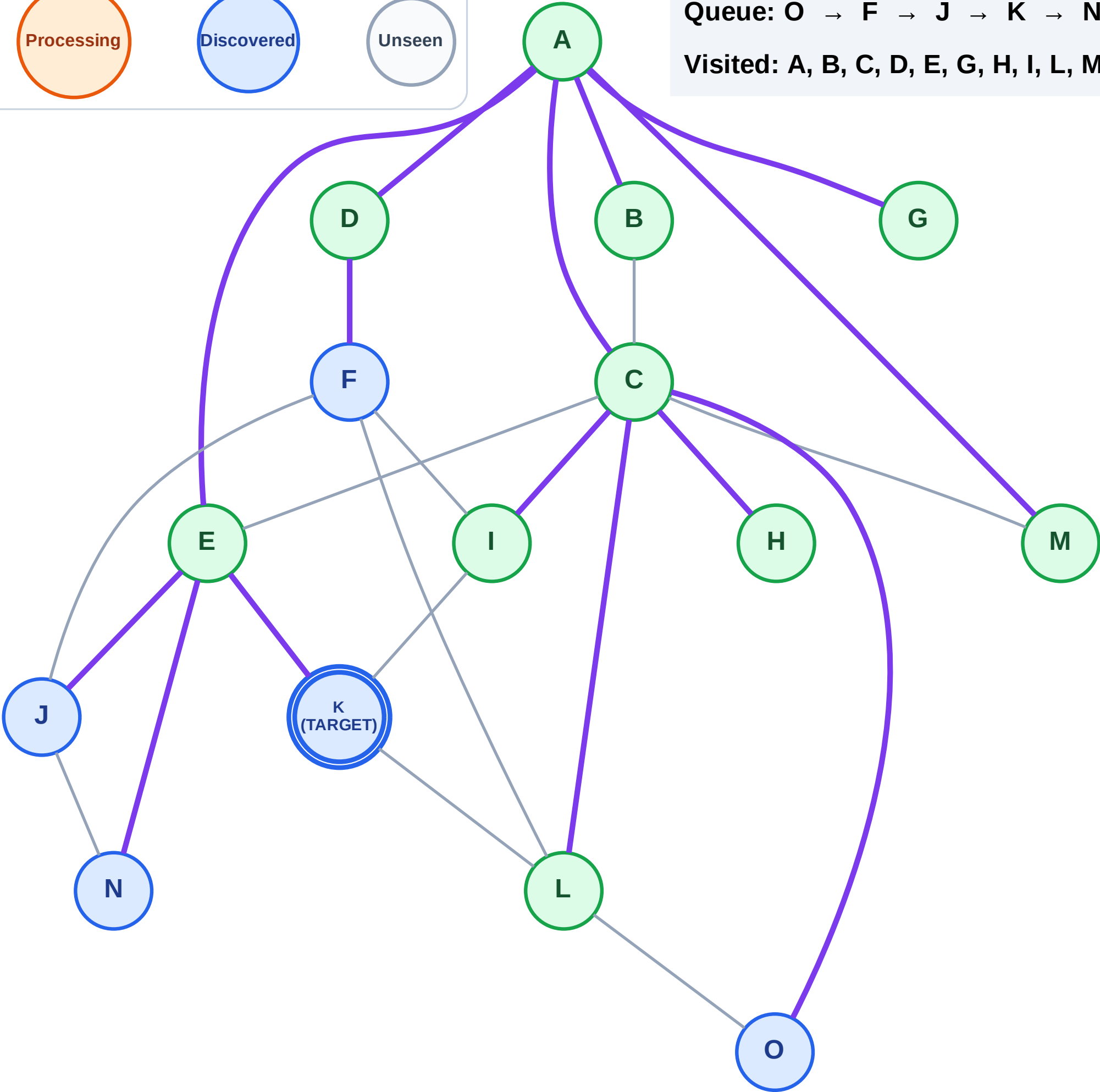
Processing

Discovered

Unseen

Queue: O → F → J → K → N

Visited: A, B, C, D, E, G, H, I, L, M



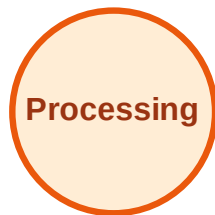
# BFS Traversal

Step 22: Process node O

## Legend



Complete



Processing



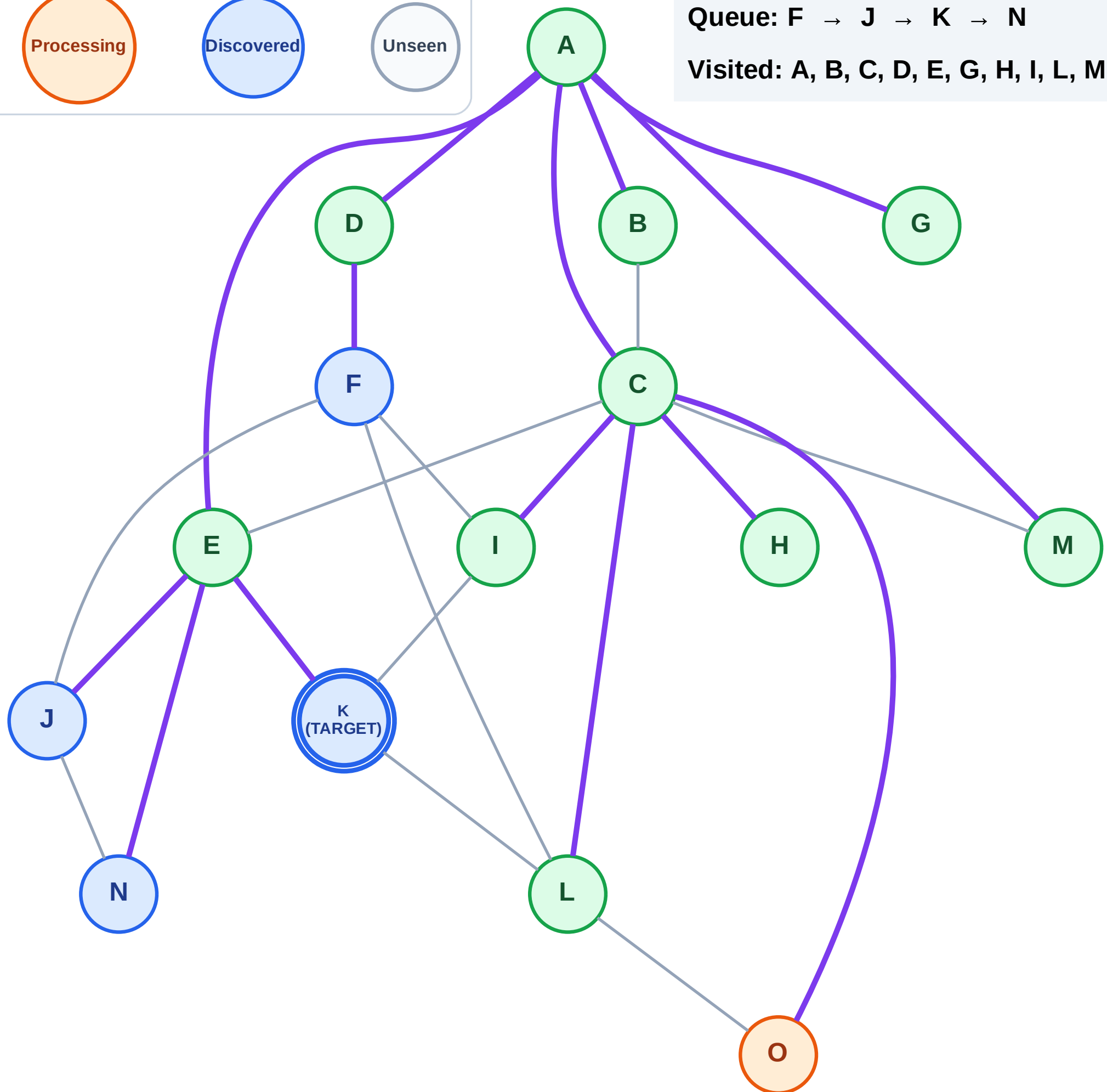
Discovered



Unseen

Queue: F → J → K → N

Visited: A, B, C, D, E, G, H, I, L, M



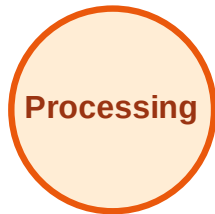
# BFS Traversal

Step 23: No new neighbors from O

## Legend



Complete



Processing



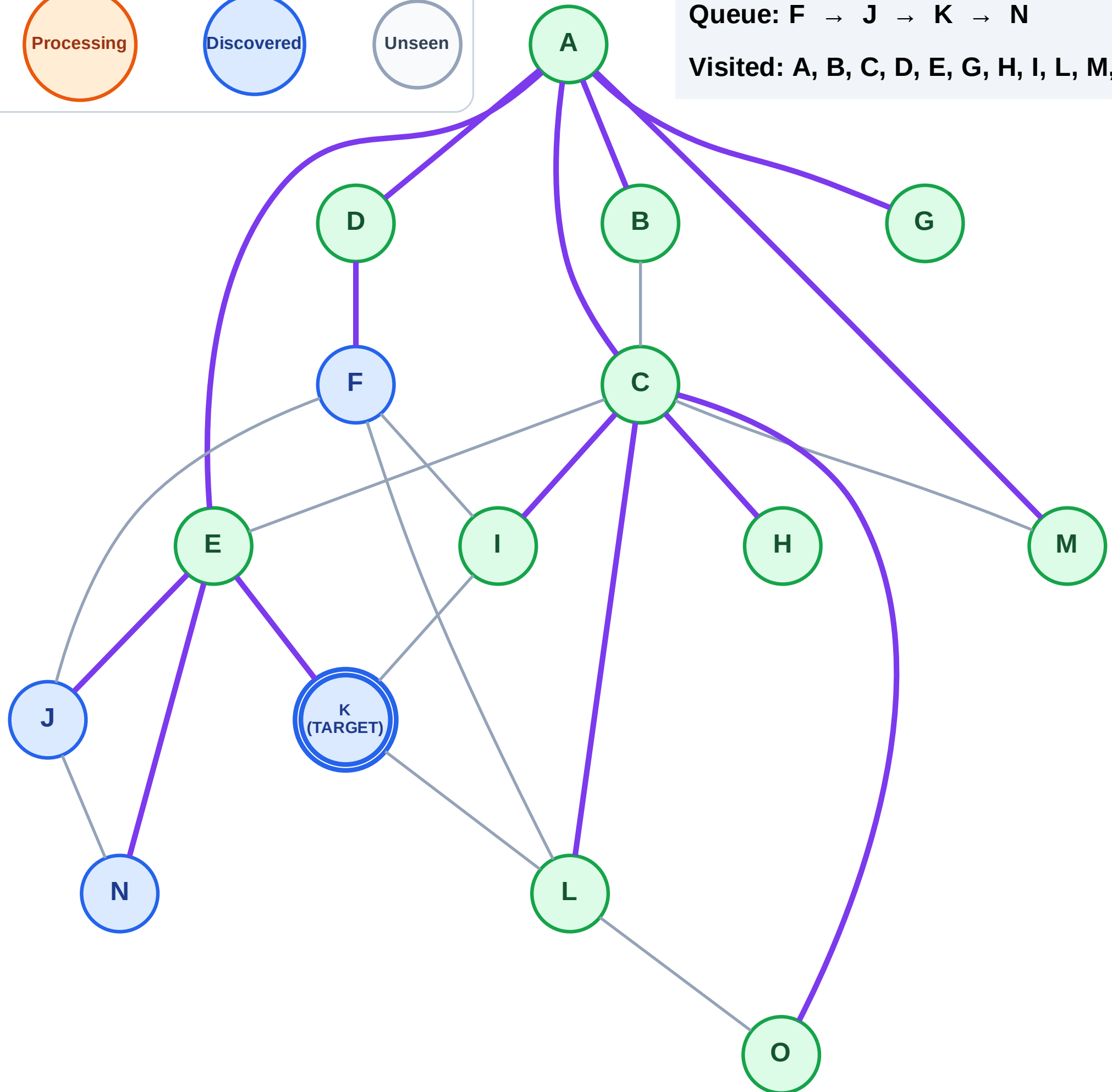
Discovered



Unseen

Queue: F → J → K → N

Visited: A, B, C, D, E, G, H, I, L, M, O



# BFS Traversal

Step 24: Process node F

Legend

Complete

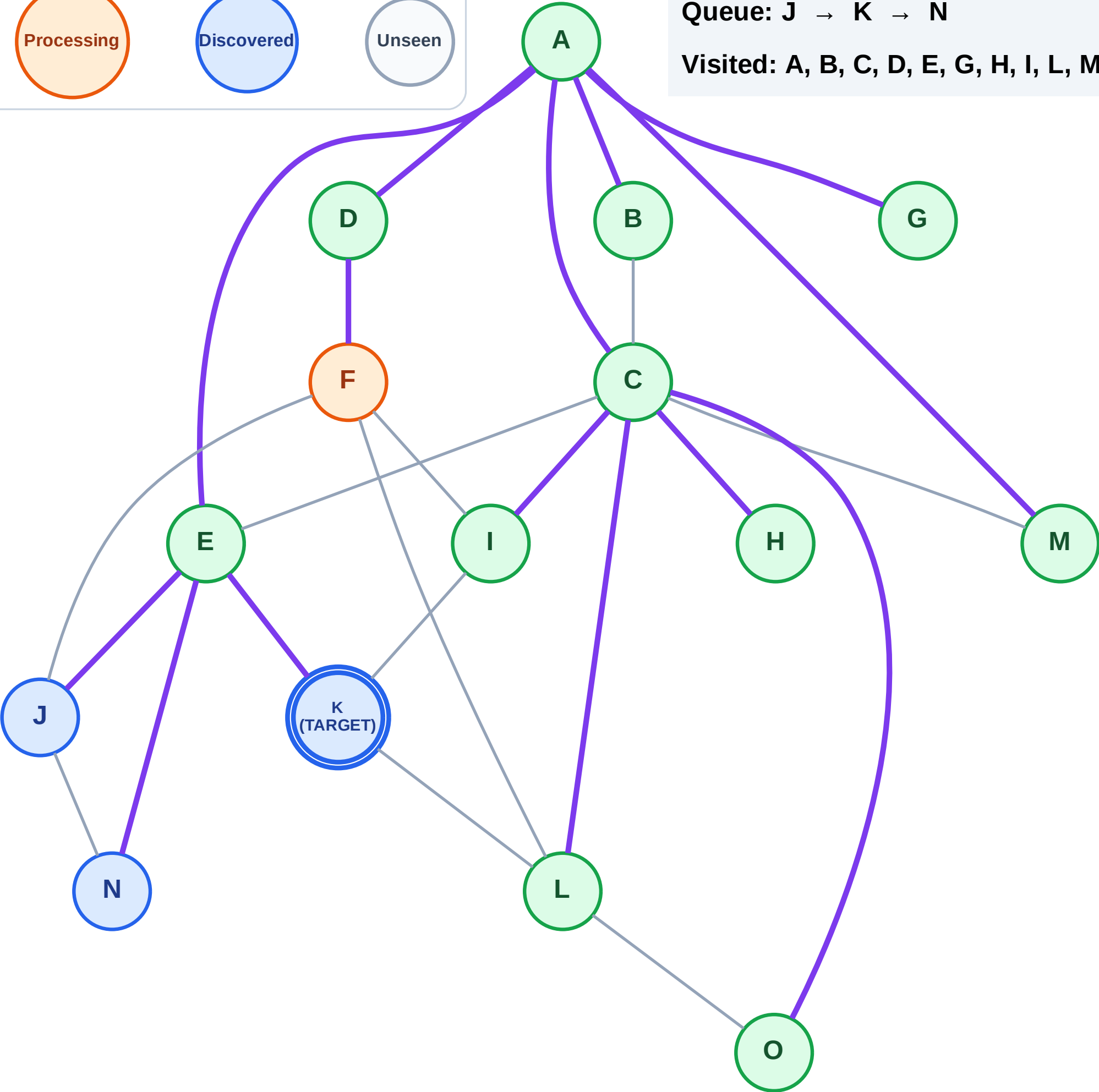
Processing

Discovered

Unseen

Queue: J → K → N

Visited: A, B, C, D, E, G, H, I, L, M, O



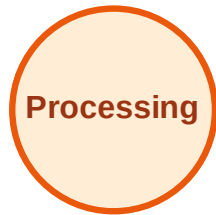
# BFS Traversal

Step 25: No new neighbors from F

## Legend



Complete



Processing



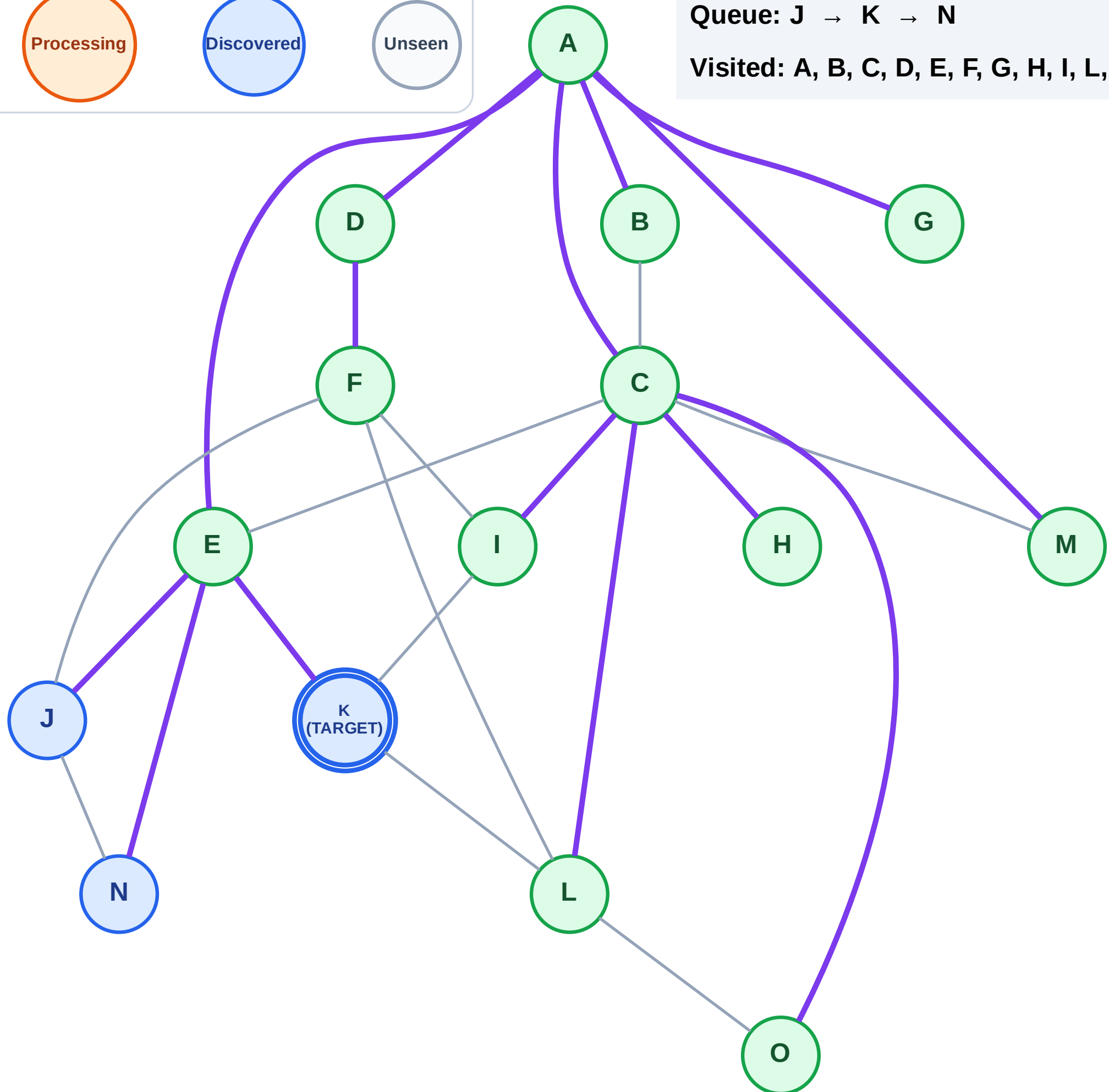
Discovered



Unseen

Queue: J → K → N

Visited: A, B, C, D, E, F, G, H, I, L, M, O



# BFS Traversal

Step 26: Process node J

Legend

Complete

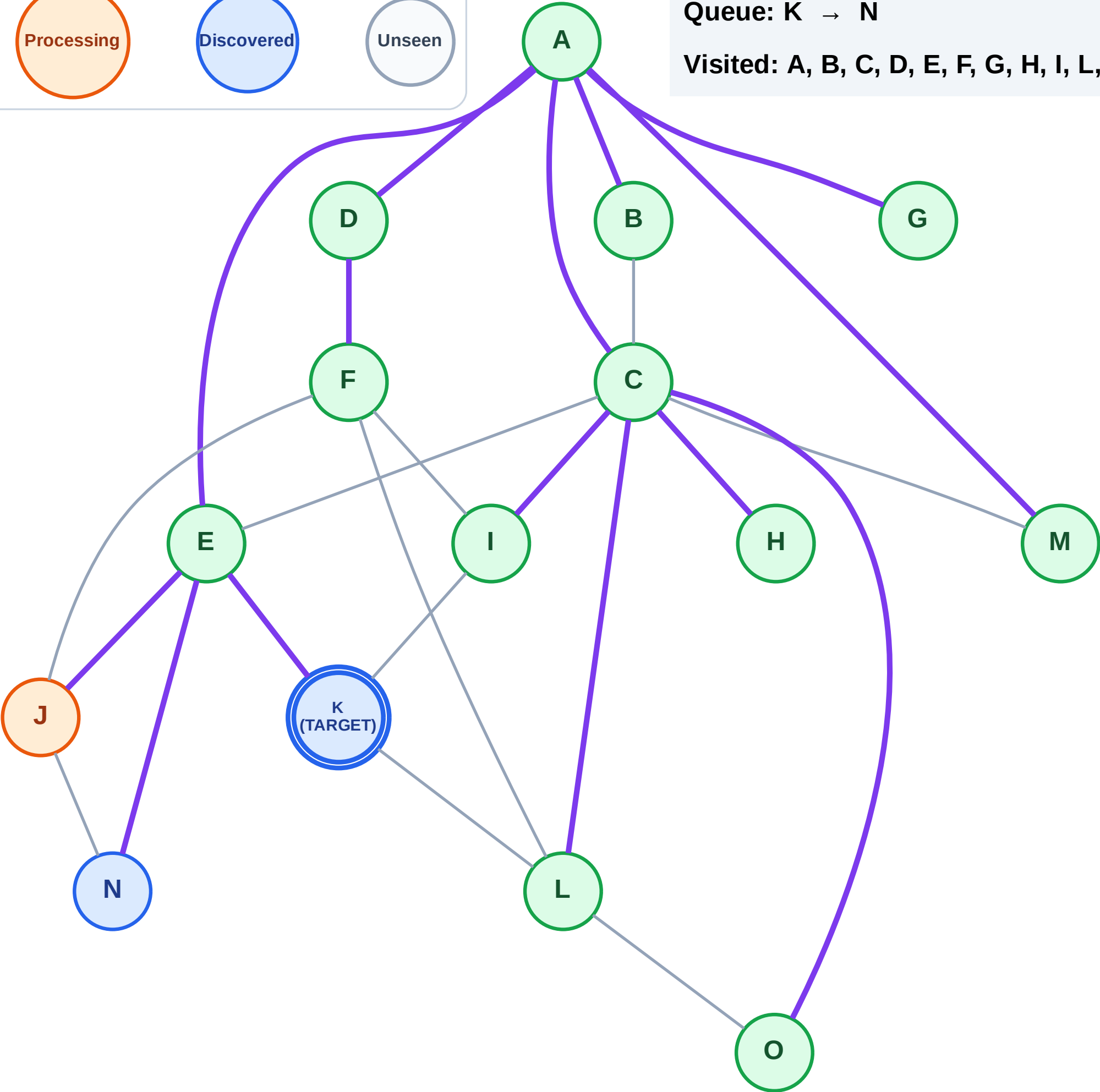
Processing

Discovered

Unseen

Queue: K → N

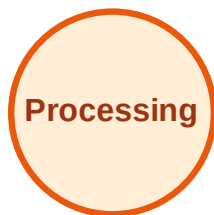
Visited: A, B, C, D, E, F, G, H, I, L, M, O



# BFS Traversal

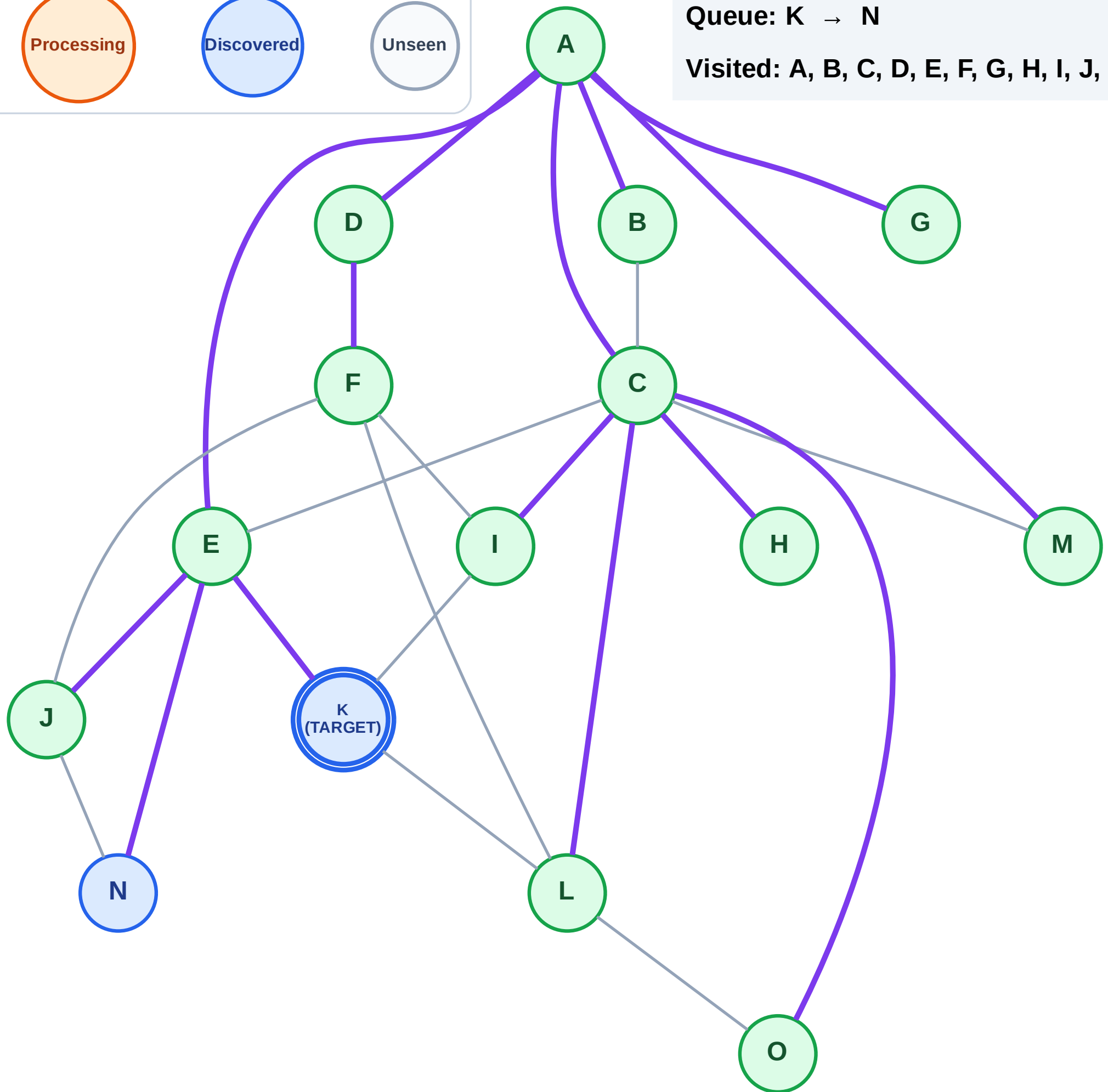
Step 27: No new neighbors from J

## Legend



Queue: K → N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



# BFS Traversal

Step 28: Process node K

## Legend

Complete

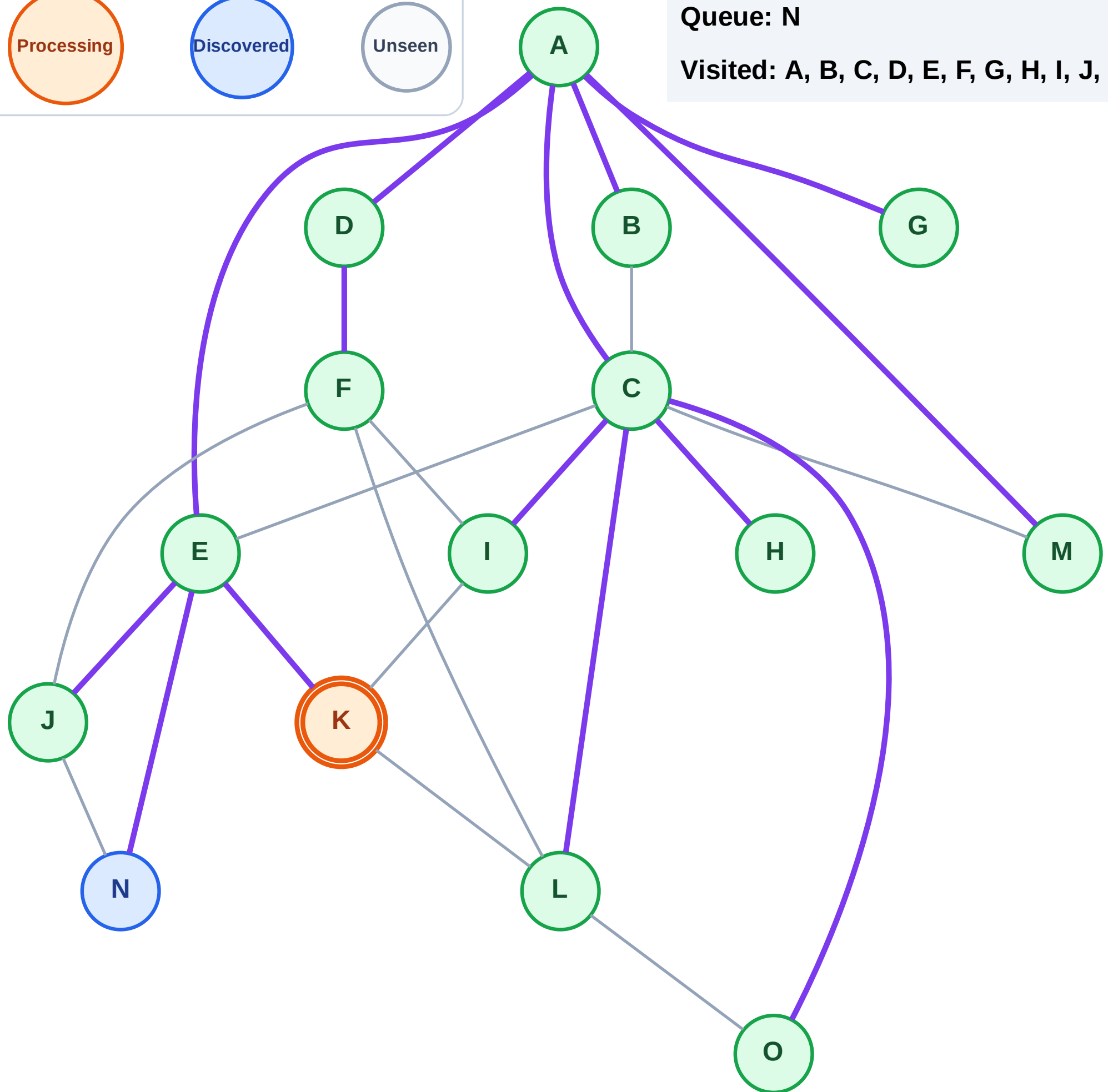
Processing

Discovered

Unseen

Queue: N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



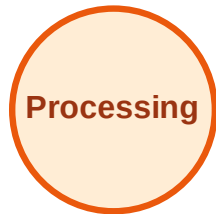
# BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → E → K

## Legend



Complete



Processing



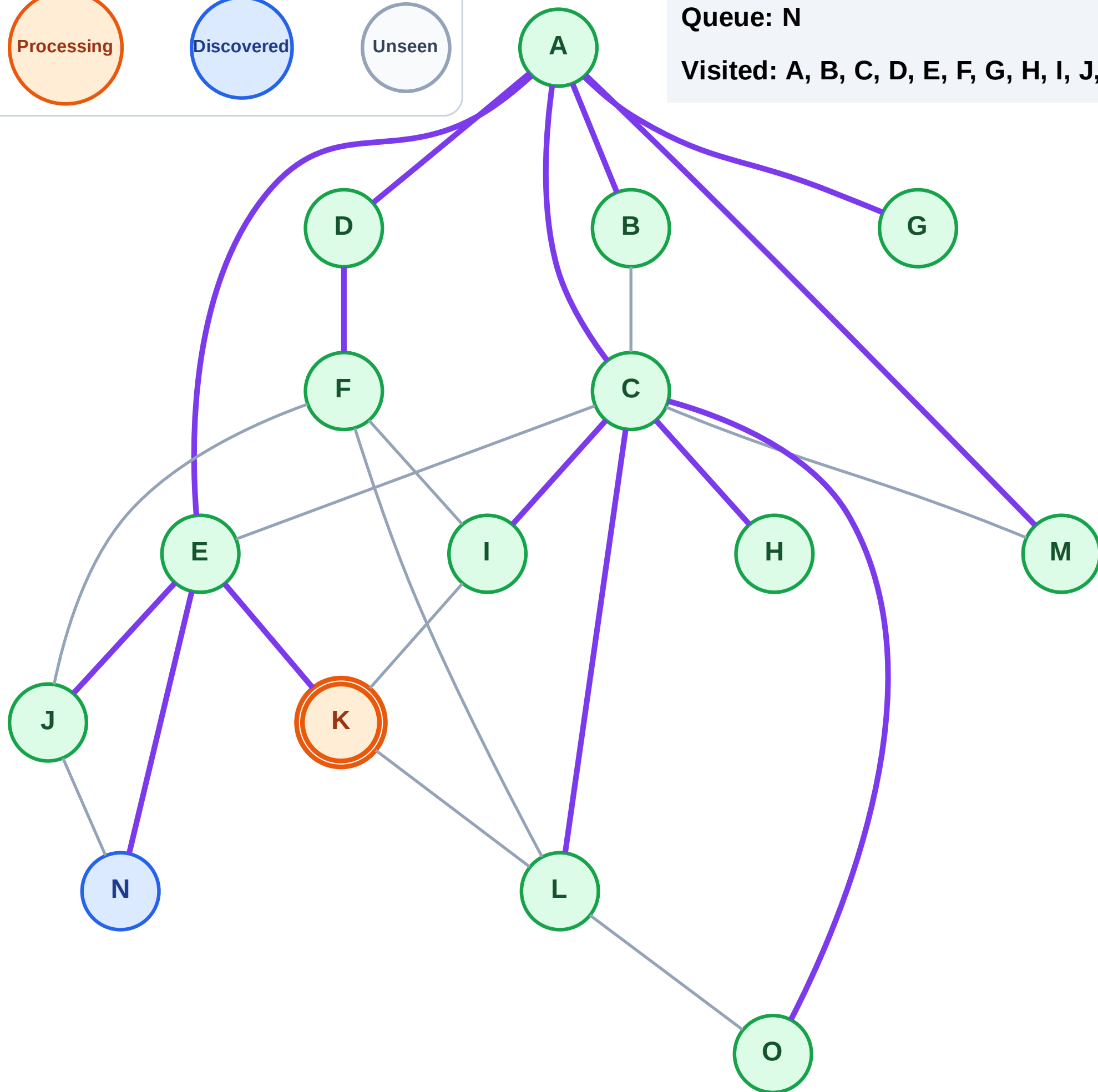
Discovered



Unseen

Queue: N

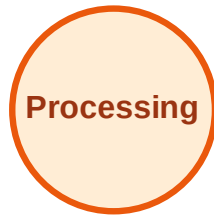
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



# DFS Traversal

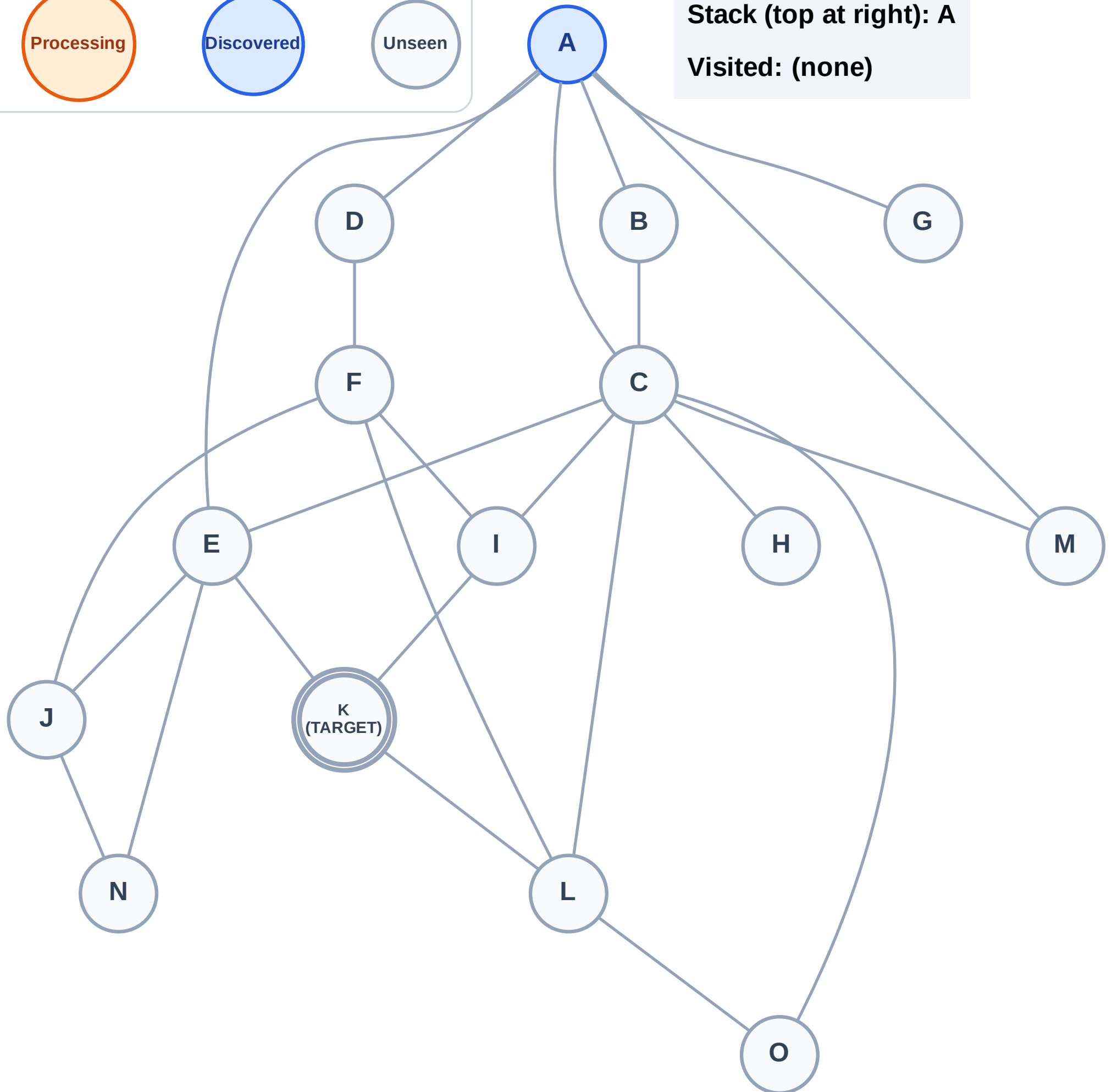
Step 1: Start at A (Seeking target node K)

## Legend



Stack (top at right): A

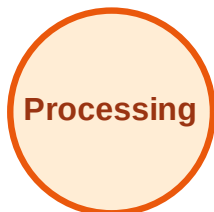
Visited: (none)



# DFS Traversal

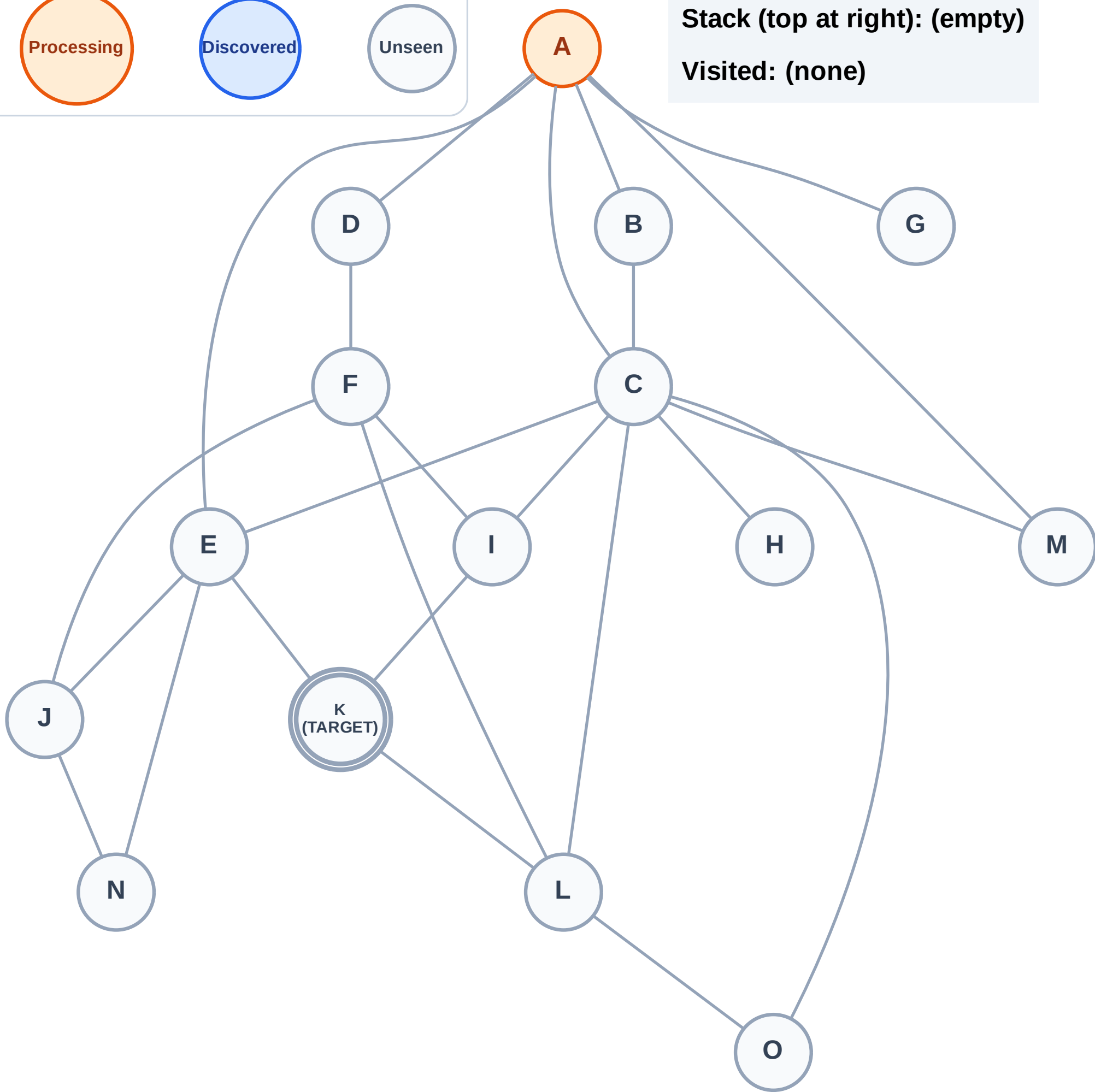
Step 2: Process node A

## Legend



Stack (top at right): (empty)

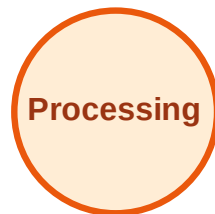
Visited: (none)



# DFS Traversal

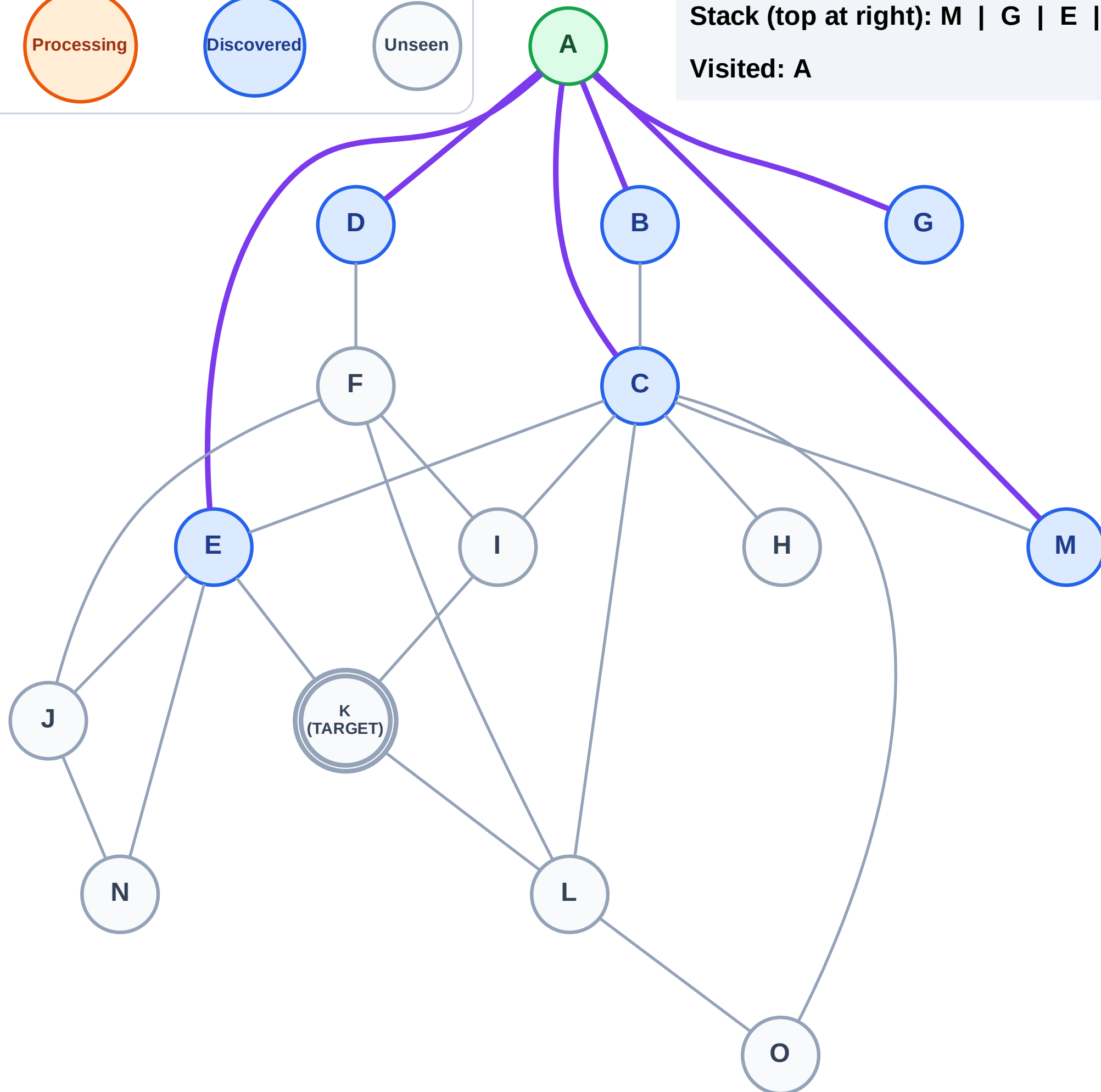
Step 3: Discover M, G, E, D, C, B from A

## Legend



Stack (top at right): M | G | E | D | C | B

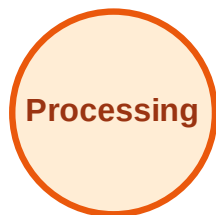
Visited: A



# DFS Traversal

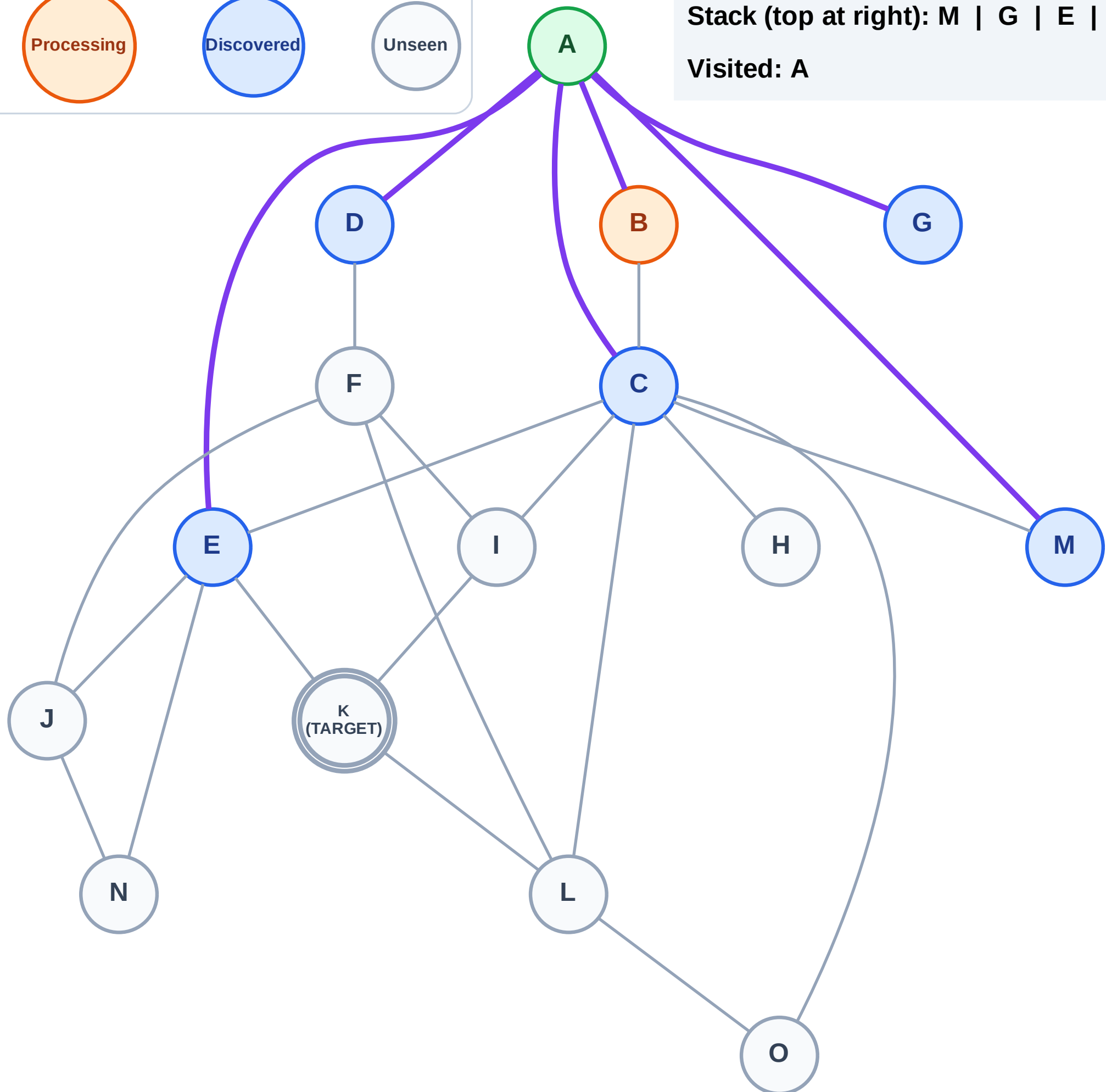
Step 4: Process node B

## Legend



Stack (top at right): M | G | E | D | C

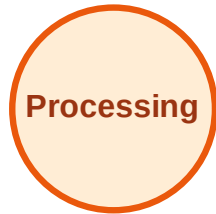
Visited: A



# DFS Traversal

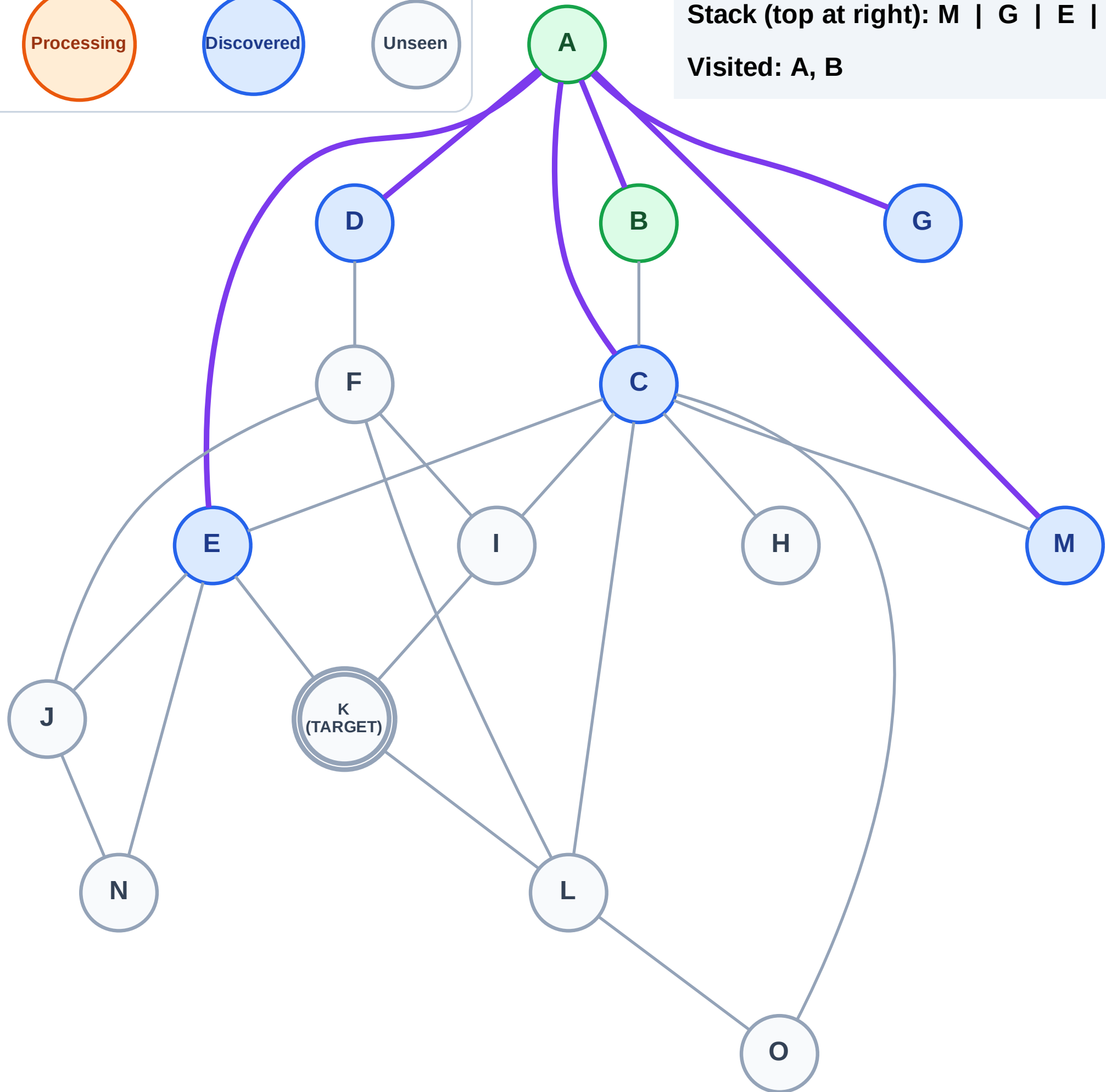
Step 5: No new neighbors from B

## Legend



Stack (top at right): M | G | E | D | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

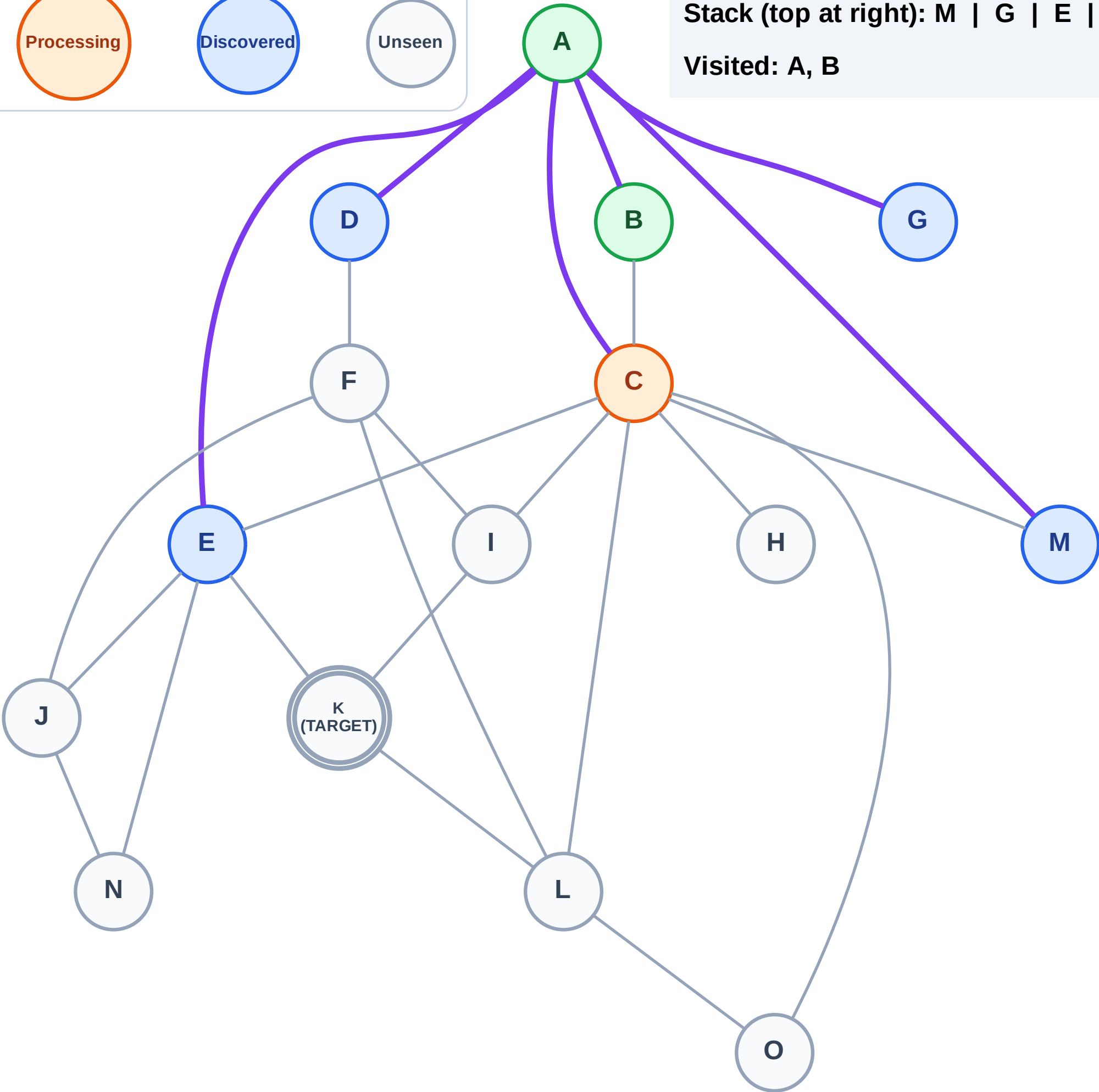
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D

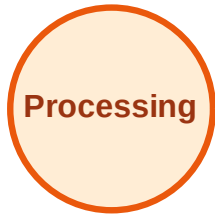
Visited: A, B



# DFS Traversal

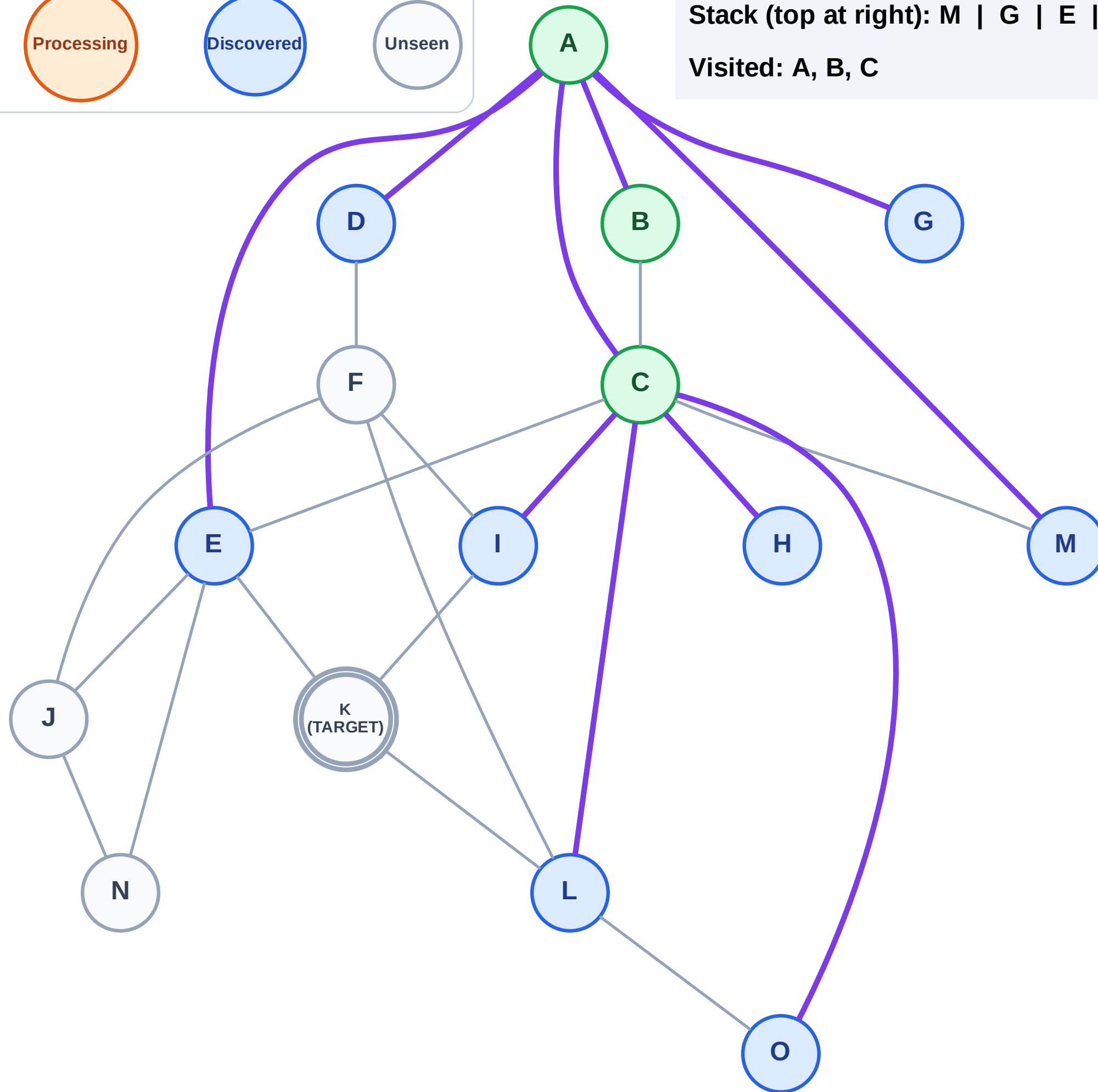
Step 7: Discover O, L, I, H from C

## Legend



Stack (top at right): M | G | E | D | O | L | I | H

Visited: A, B, C



# DFS Traversal

Step 8: Process node H

Legend

Complete

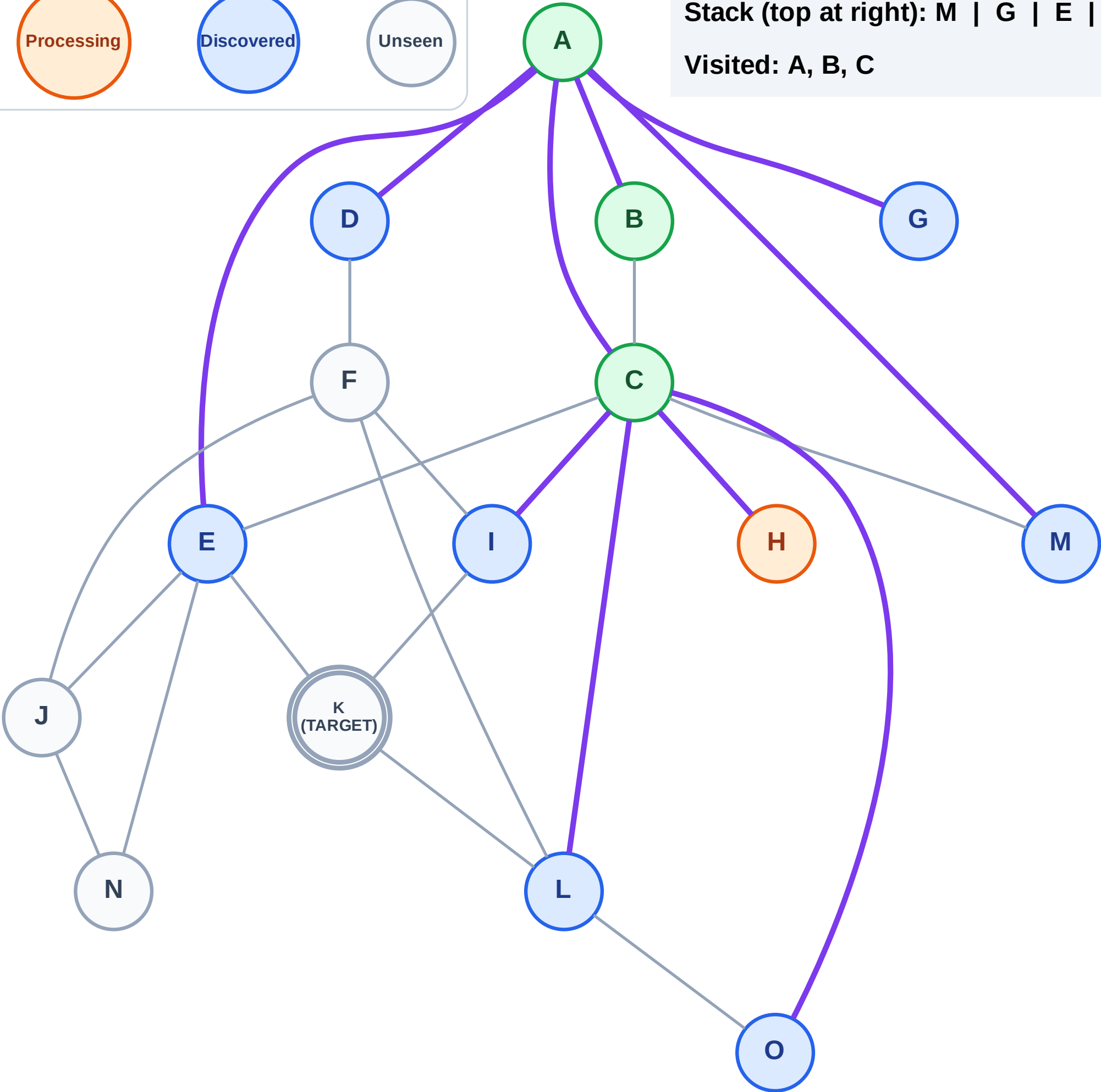
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | I

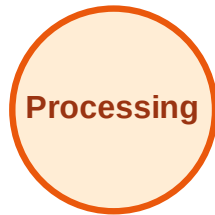
Visited: A, B, C



# DFS Traversal

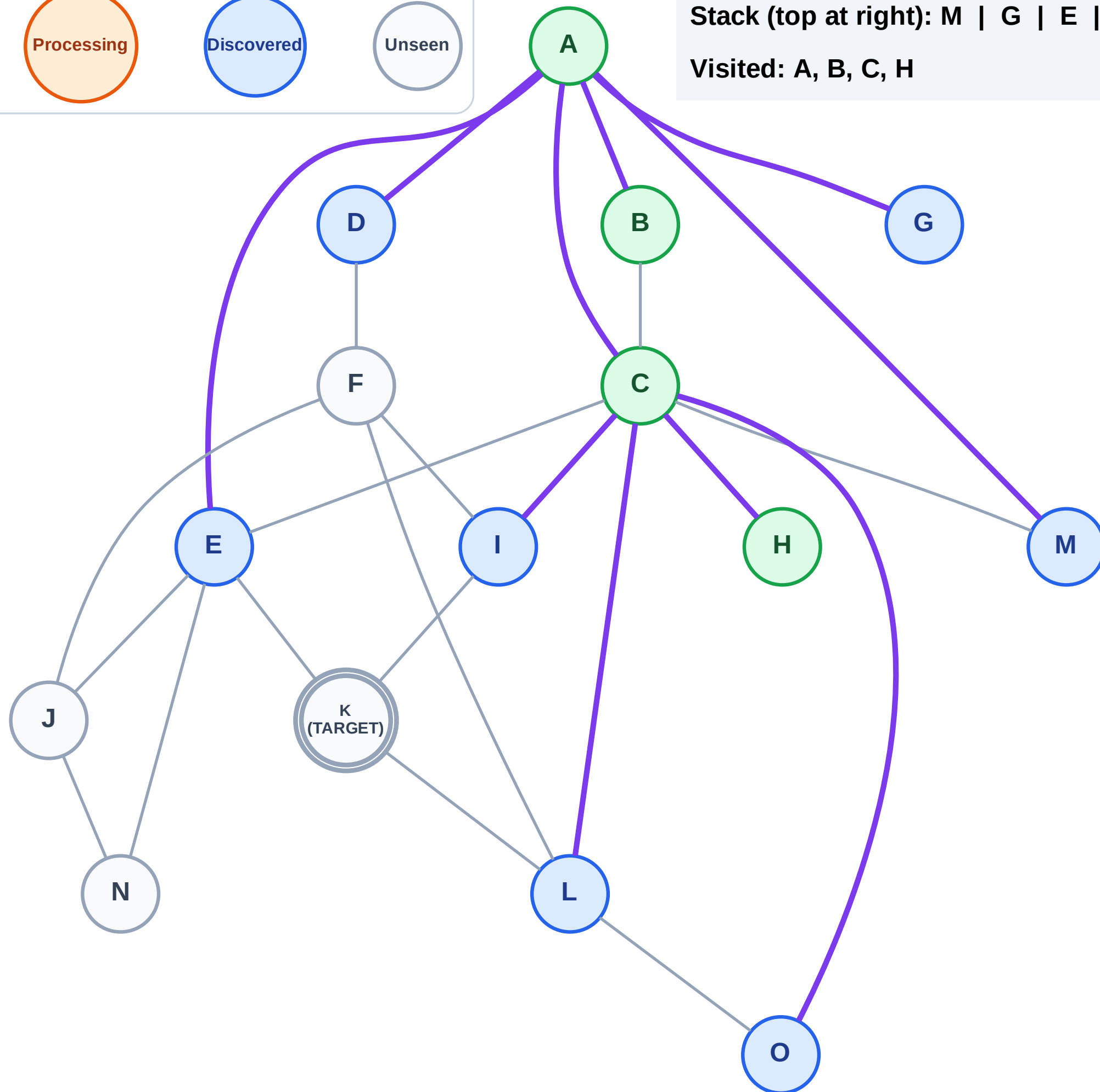
Step 9: No new neighbors from H

## Legend



Stack (top at right): M | G | E | D | O | L | I

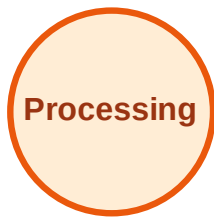
Visited: A, B, C, H



# DFS Traversal

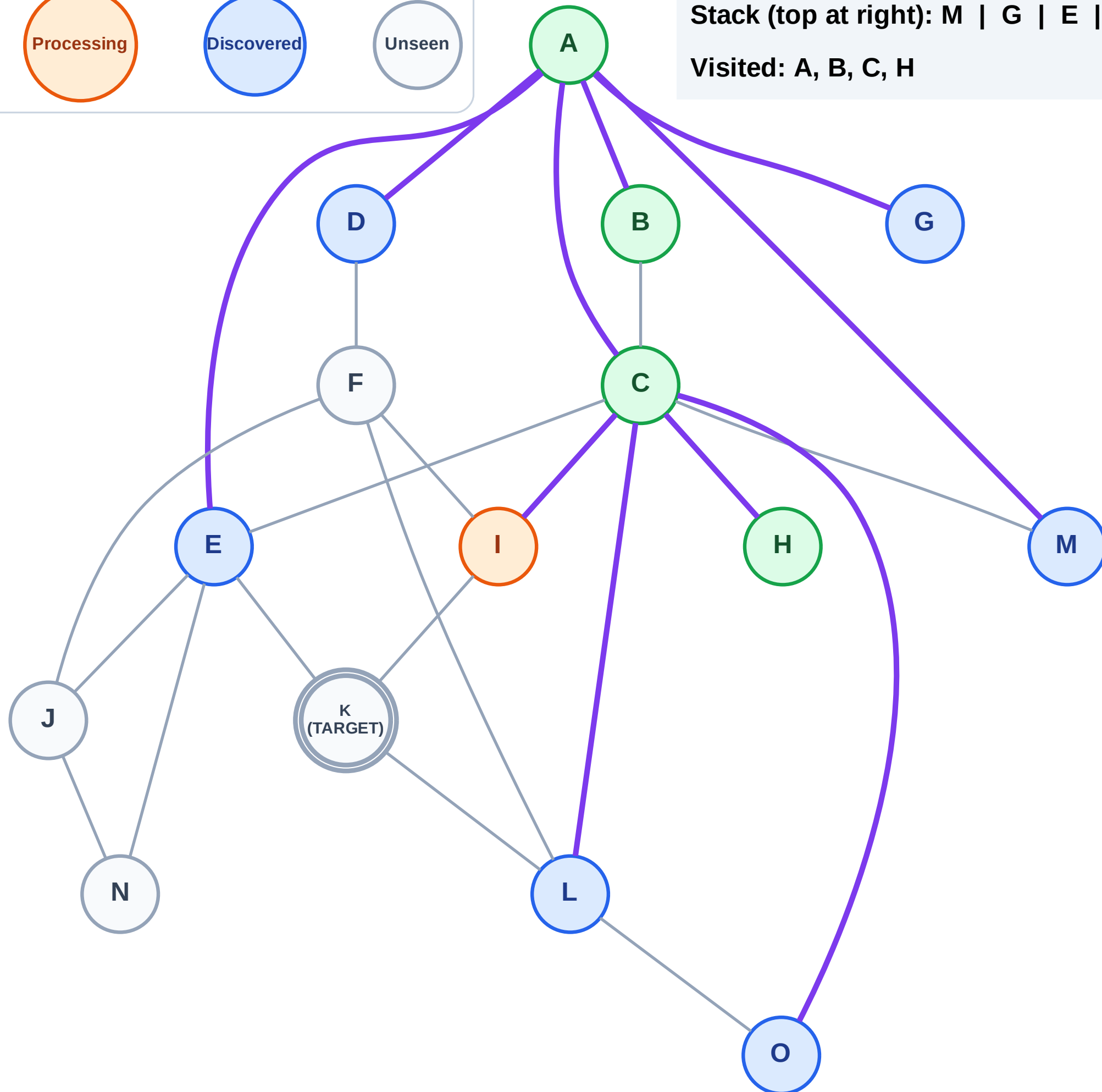
Step 10: Process node I

## Legend



Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, H



# DFS Traversal

Step 11: Discover K, F from I

Legend

Complete

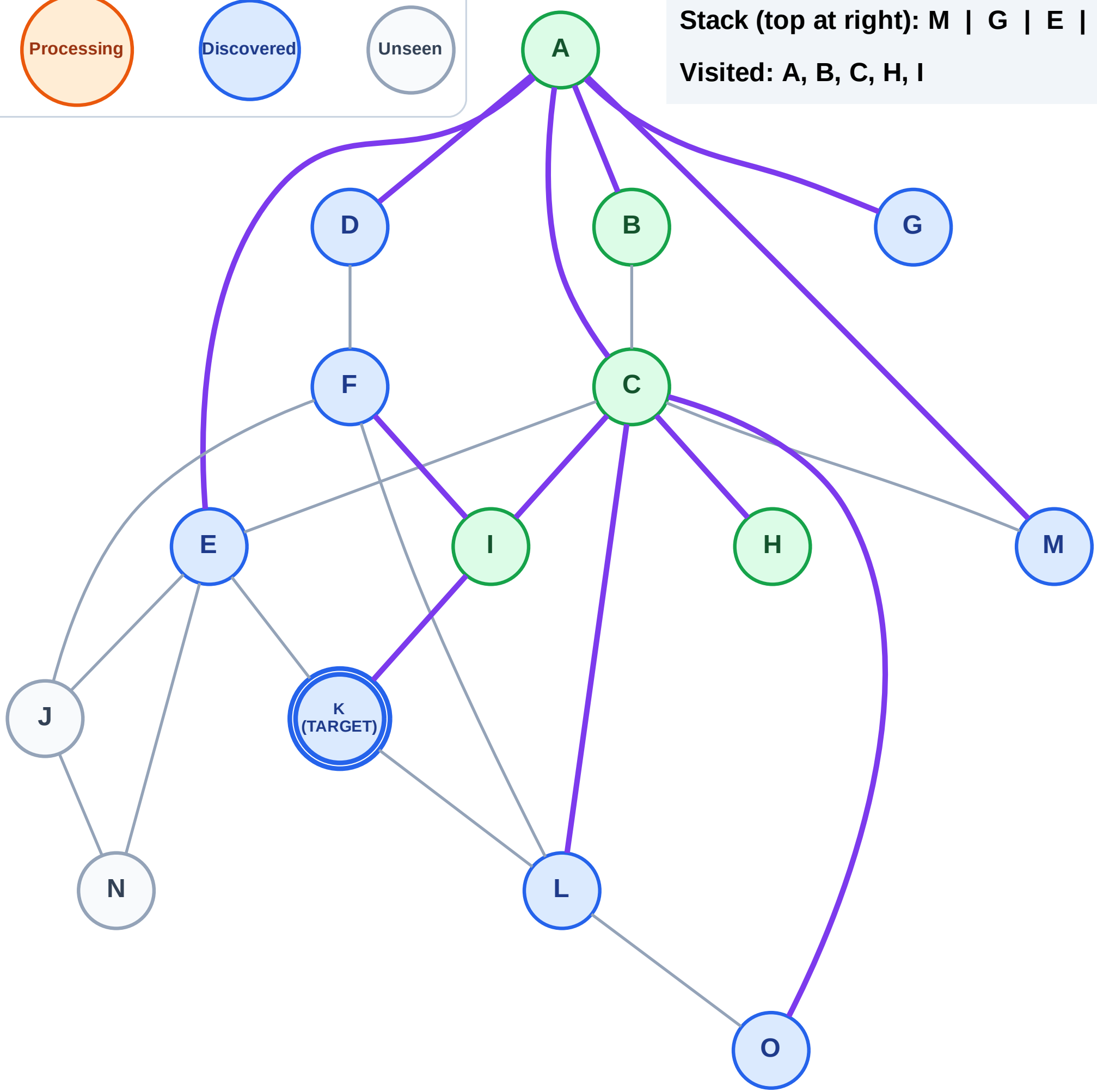
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | K | F

Visited: A, B, C, H, I



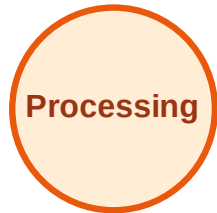
# DFS Traversal

Step 12: Process node F

## Legend



Complete



Processing



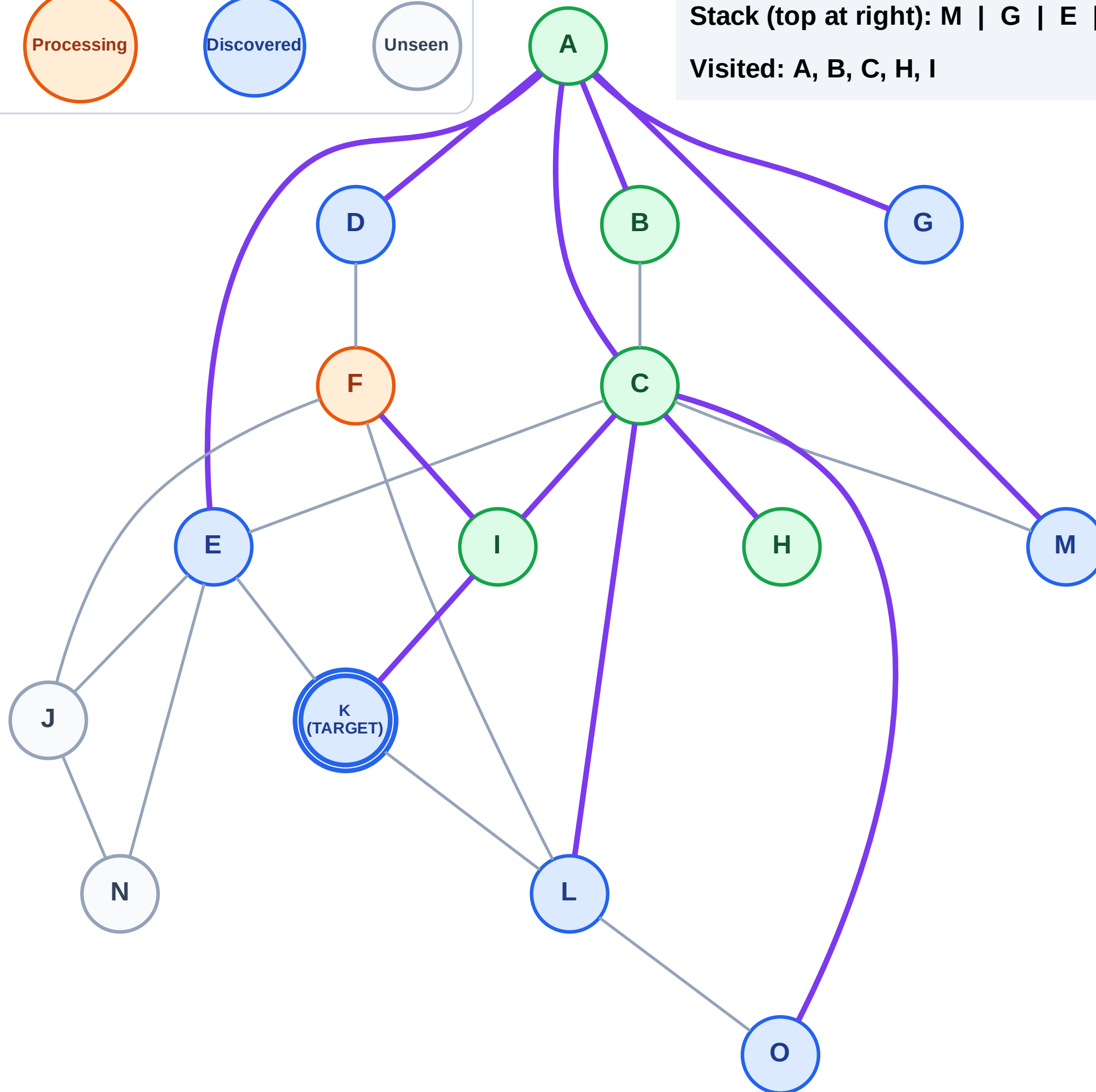
Discovered



Unseen

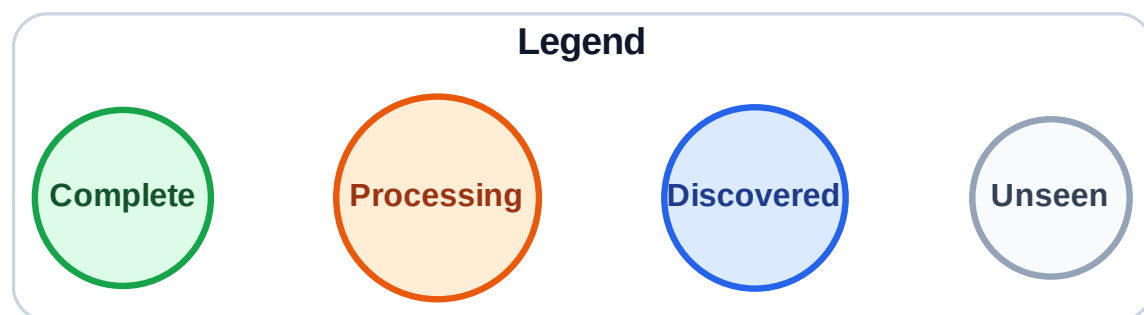
Stack (top at right): M | G | E | D | O | L | K

Visited: A, B, C, H, I



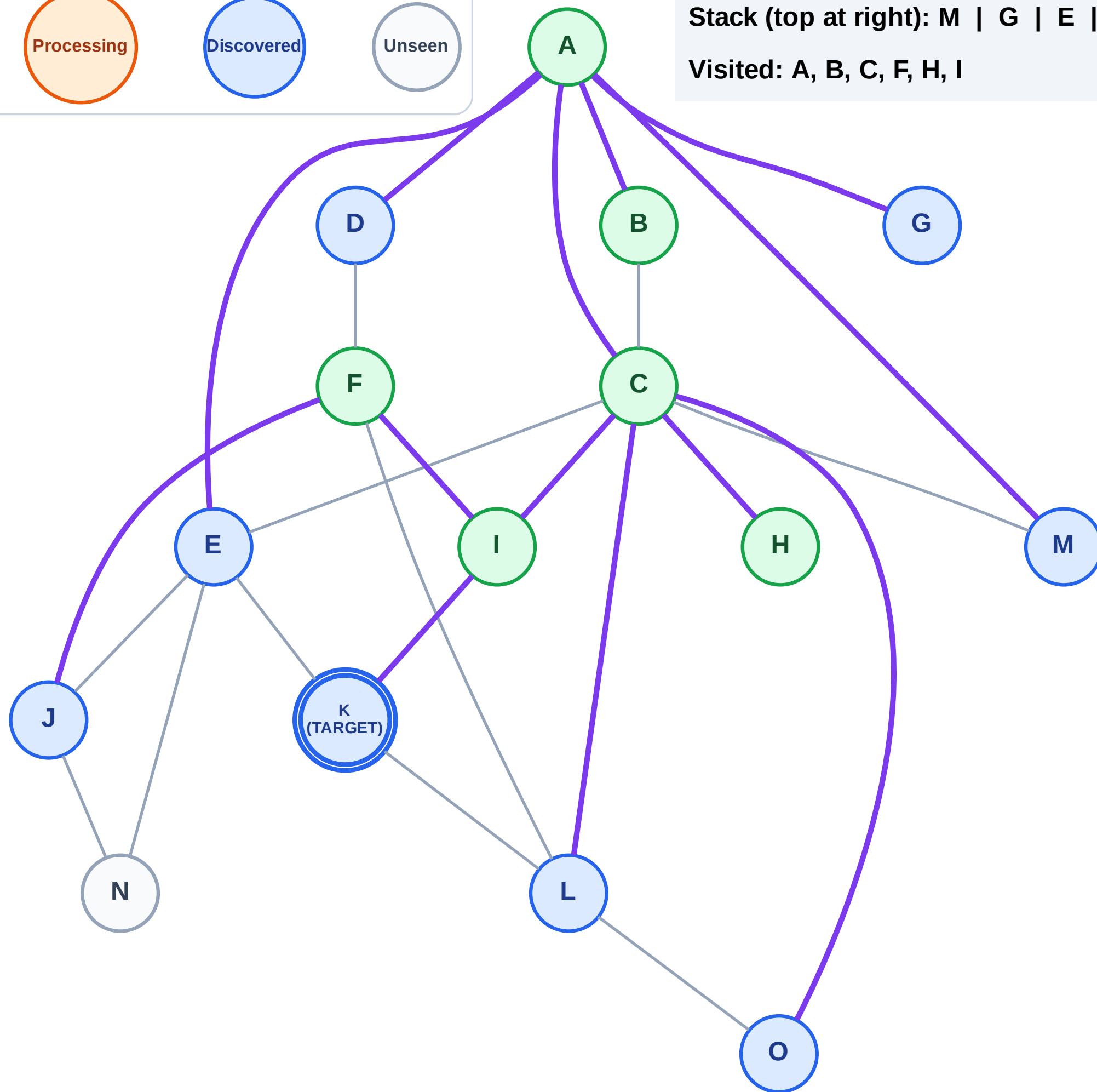
# DFS Traversal

Step 13: Discover J from F



Stack (top at right): M | G | E | D | O | L | K | J

Visited: A, B, C, F, H, I



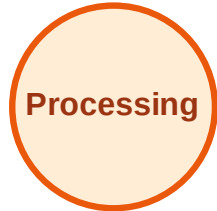
# DFS Traversal

Step 14: Process node J

## Legend



Complete



Processing



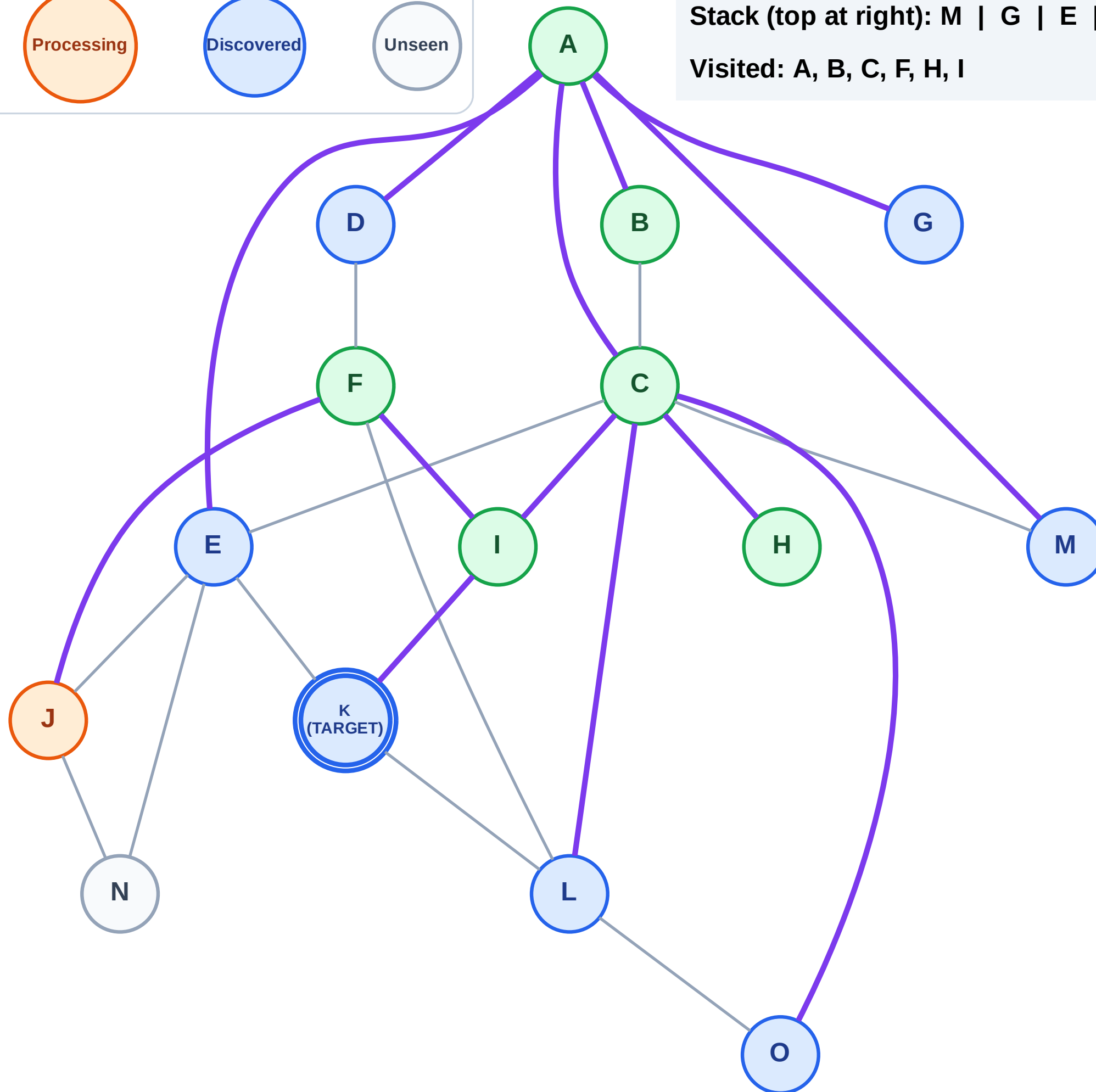
Discovered



Unseen

Stack (top at right): M | G | E | D | O | L | K

Visited: A, B, C, F, H, I



# DFS Traversal

Step 15: Discover N from J

Legend

Complete

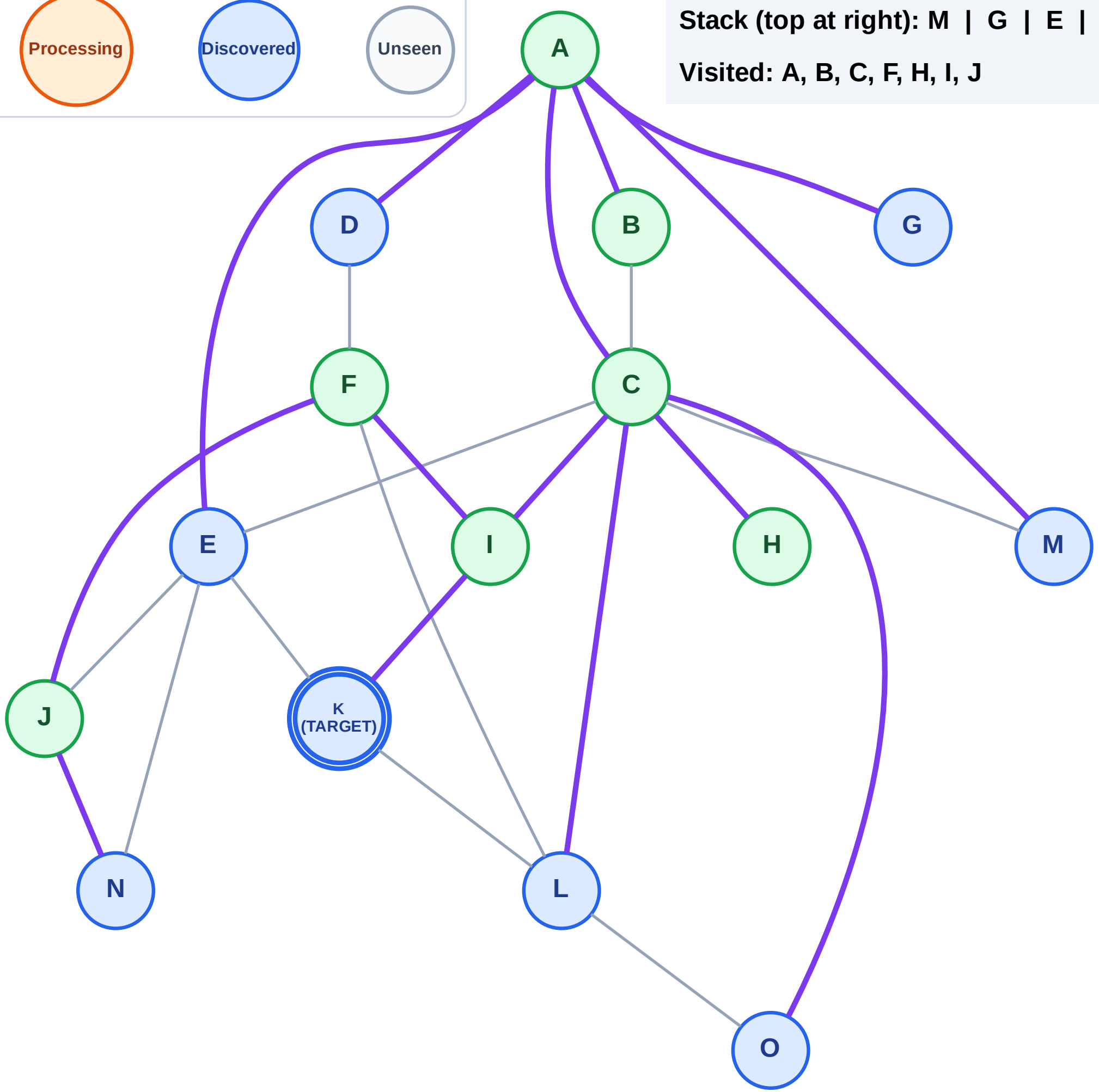
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | K | N

Visited: A, B, C, F, H, I, J



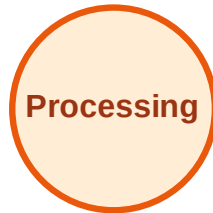
# DFS Traversal

Step 16: Process node N

## Legend



Complete



Processing



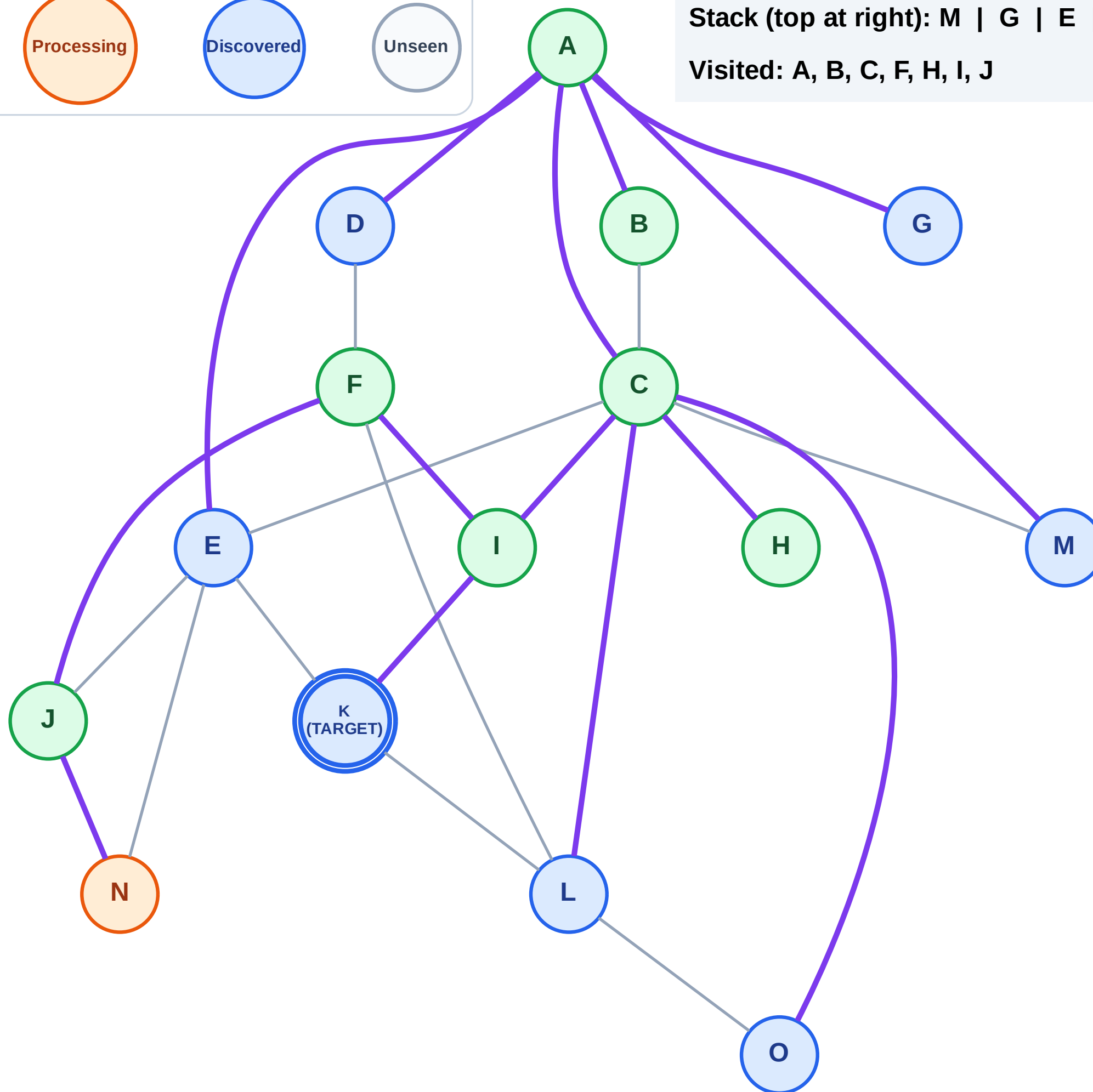
Discovered



Unseen

Stack (top at right): M | G | E | D | O | L | K

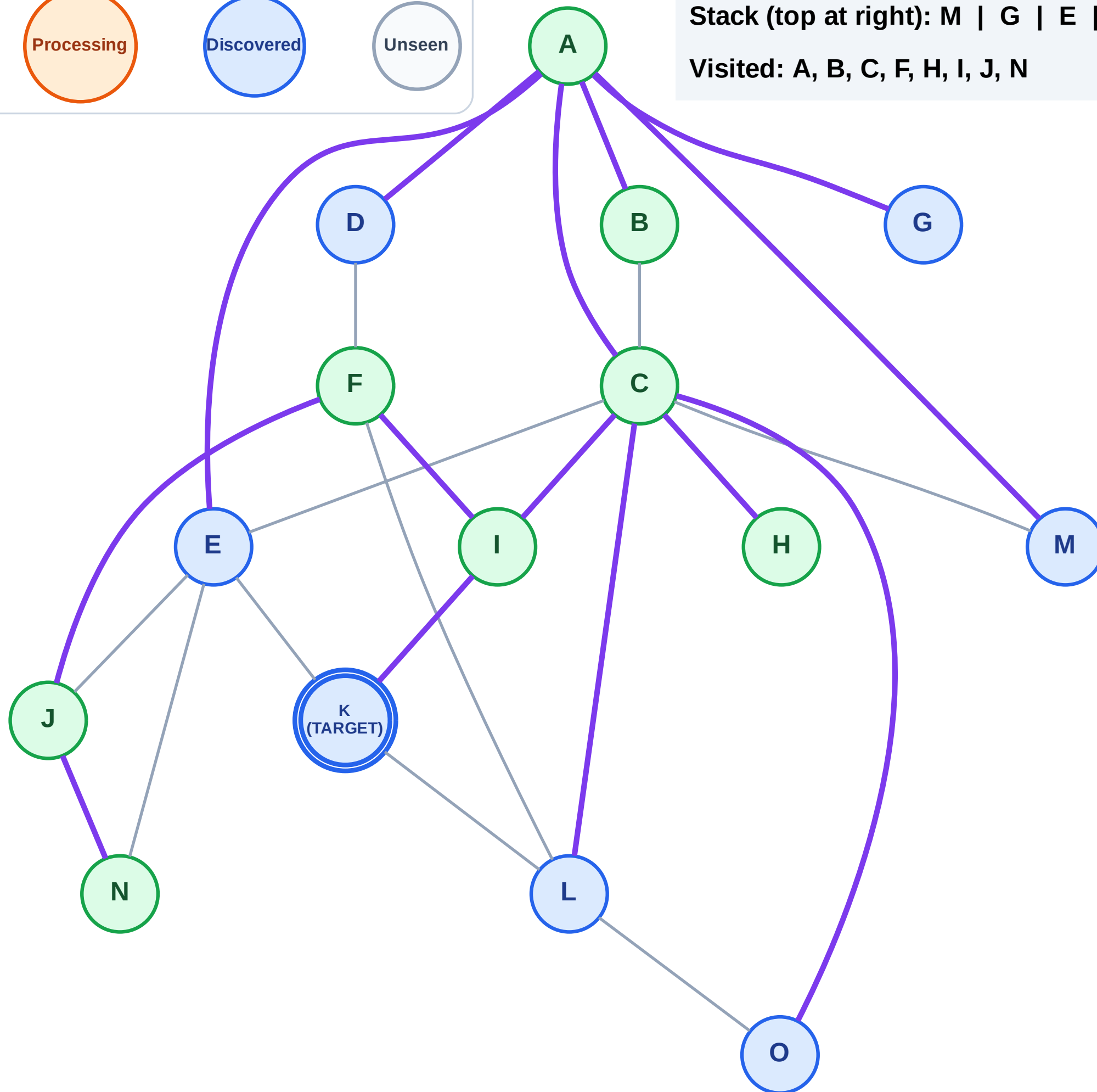
Visited: A, B, C, F, H, I, J



Step 17: No new neighbors from N

**Stack (top at right):** M | G | E | D | O | L | K

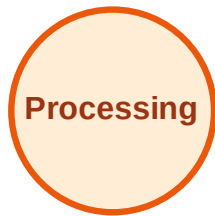
**Visited:** A, B, C, F, H, I, J, N



# DFS Traversal

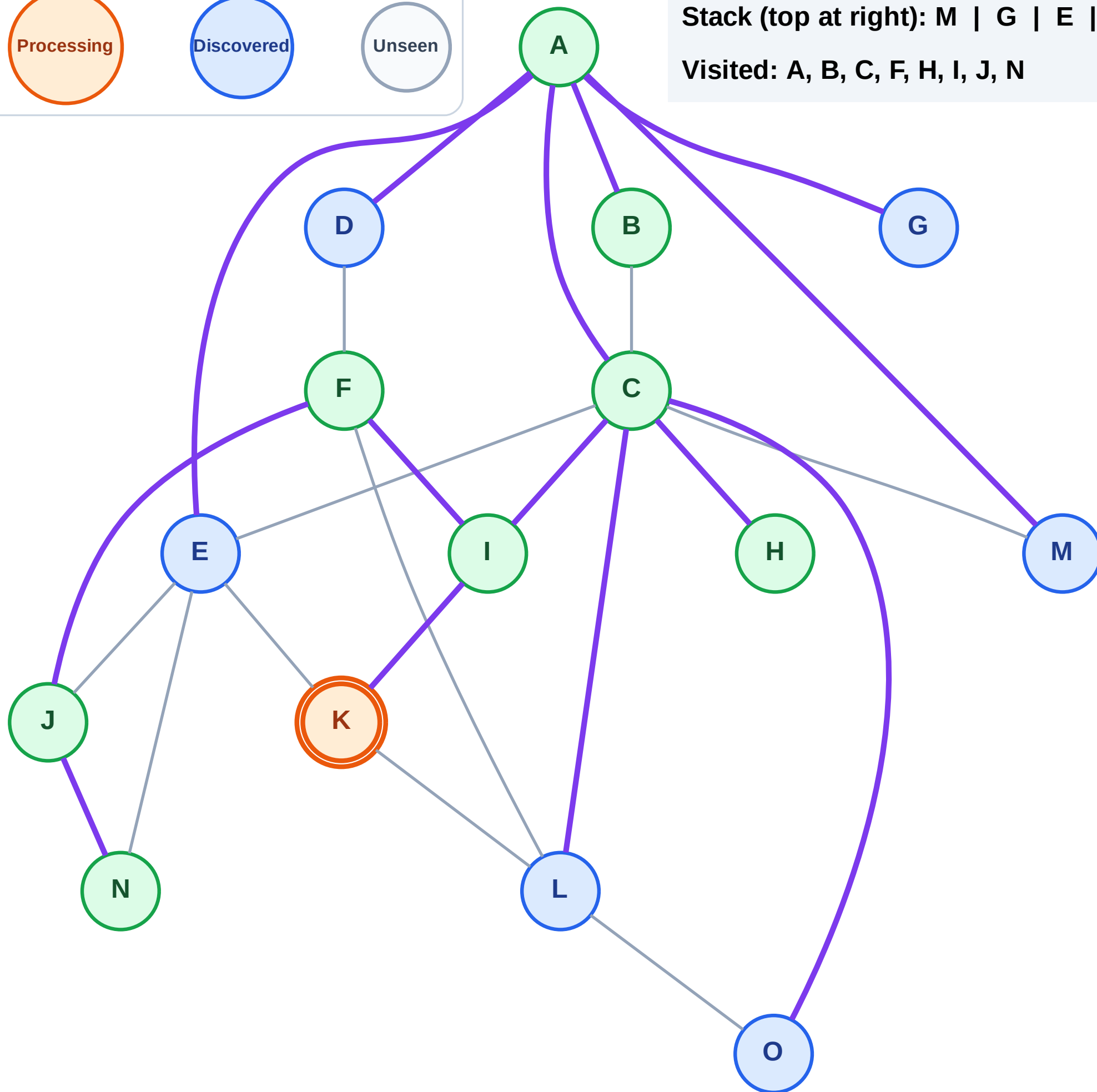
Step 18: Process node K

## Legend



Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, F, H, I, J, N



# DFS Traversal

Step 19: TARGET NODE K REACHED! Path (3 hops): A → C → I → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, F, H, I, J, K, N

